# Commutative Algebra

Shing Tak Lam

Michaelmas 2023*

## Contents

**1 Chain conditions** 2

**2 Tensor products** 4
2.1 Tensor product of $R$-linear maps . . . 9
2.2 Tensor product of algebras . . . 10
2.3 Restriction and extension of scalars . . . 12
2.4 Exactness properties of the tensor product . . . 16
2.5 Flat modules – a first encounter . . . 18
2.6 Further examples of tensor products . . . 20

**3 Localisation** 22
3.1 Construction . . . 22
3.2 Extension and contraction of ideals . . . 28
3.3 Local properties . . . 31
3.4 Localisation as a quotient . . . 33

**4 Nakayama's lemma** 33

**5 Integral and finite extensions** 35

**6 Noether normalisation and Hilbert's Nullstellensatz** 39
6.1 Noether normalisation . . . 39
6.2 Hilbert Nullstellensatz . . . 41

**7 Integral and finite extensions again** 43
7.1 Cohen-Seidenberg theorems . . . 45

**8 Primary decomposition** 47

Lecture 1

In this course, a ring is a commutative unital ring $R$. One non-commutative exception is the ring $\mathrm{End}(M)$, where $M$ is an abelian group. This is a ring with pointwise addition, and composition as multiplication.

**Definition 0.0.1** (module)

An *$R$-module* $M$ is an abelian group $M$ with an fixed ring homomorphism $\rho : R \to \mathrm{End}(M)$. We will write $r \cdot m := \rho(r)(m)$.

**Remark 0.0.2.** By definition, this implies that $(r_1 + r_2) \cdot m = r_1 \cdot m + r_2 \cdot m$, $r \cdot (m_1 + m_2) = r \cdot m_1 + r \cdot m_2$ and $r_1 \cdot (r_2 \cdot m) = (r_1 r_2) \cdot m$.

*Based on lectures by Oren Becker. Last updated November 14, 2023.

**Example 0.0.3** (Examples of modules)

- Let $k$ be a field. Then a $k$-module is the same as a $k$-vector space.
- Every abelian group is a $\mathbb{Z}$-module in a unique way, since we must have that $\rho(1) = \mathrm{id}_M$. Therefore, abelian groups and $\mathbb{Z}$-modules are the same thing.
- Every ring $R$ is (trivially) an $R$-module.
- More generally, $R^{\oplus\mathbb{N}}$ (direct sum) and $R^{\mathbb{N}}$ (direct product) are $R$-modules.

# 1 Chain conditions

**Definition 1.0.1** (Noetherian, Artinian module)

An $R$-module $M$ is *Noetherian* if one of the following (equivalent) conditions hold:

1. Every ascending chain of submodules $M_0 \subseteq M_1 \subseteq M_2 \subseteq \cdots$ stabilises. That is, it is eventually constant.
2. Every non-empty set $\Sigma$ of submodules of $M$ has a maximal element.

$M$ is *Artinian* if we replace in the above: ascending with descending, maximal with minimal.

**Lemma 1.0.2.** An $R$-module $M$ is Noetherian if and only if every submodule of $M$ is finitely generated.

In particular, every Noetherian module is finitely generated. If $R = \mathbb{Z}[T_1, T_2, \dots]$, with $M = R$ as an $R$-module. Then $M$ is finitely generated. On the other hand, $M' = \langle T_1, T_2, T_3, \dots \rangle$, is not finitely generated.

**Definition 1.0.3** (Noetherian, Artinian ring)

A ring $R$ is Noetherian (resp. Artinian) if it is Noetherian (resp. Artinian) as an $R$-module.

**Example 1.0.4**

1. $\mathbb{Z}$ is Noetherian (as it is a PID), but not Artinian (e.g. $\langle 2 \rangle \supseteq \langle 4 \rangle \supseteq \langle 8 \rangle \supseteq \cdots$).
2. $\mathbb{Z}[1/2]/\mathbb{Z}$ is Artinian, but not Noetherian as a $\mathbb{Z}$-module.
3. A ring $R$ is Artinian if and only if $R$ is Noetherian and $R$ has Krull dimension 0.

**Definition 1.0.5** (Exact sequence)

A sequence

$$\cdots \longrightarrow M_{i-1} \xrightarrow{f_i} M_i \xrightarrow{f_{i+1}} M_{i+1} \longrightarrow \cdots$$

of $R$-modules and $R$-module homomorphisms is *exact* if $\mathrm{im}(f_i) = \ker(f_{i+1})$ for all $i$.

**Definition 1.0.6** (Short exact sequence)

A *short exact sequence* (SES) is an exact sequence of the form

$$0 \longrightarrow N \xhookrightarrow{\iota} M \twoheadrightarrow L \longrightarrow 0$$

That is, we have an embedding $\iota : N \hookrightarrow M$, and an isomorphism $L \cong M/\iota(N)$.

**Lemma 1.0.7.** Let

$$0 \longrightarrow N \longrightarrow M \longrightarrow L \longrightarrow 0$$

be an SES of $R$-modules. Then $M$ is Noetherian (resp. Artinian) if and only if $N$ and $L$ are Noetherian (resp. Artinian).

*Proof.* We may assume without loss of generality that $N$ is a submodule of $M$. Let $P_1 \subseteq P_2 \subseteq \dots$ be an increasing (resp. decreasing) sequence of submodules of $M$. In this case,

$$N \cap P_1 \subseteq N \cap P_2 \subseteq \cdots$$

is an increasing (resp. decreasing) sequence of submodules of $N$, hence eventually constant. Similarly,

$$\frac{N + P_1}{N} \subseteq \frac{N + P_2}{N} \subseteq \cdots$$

is an increasing (resp. decreasing) sequence of submodules of $L = M/N$, hence eventually constant. For large $n$, we will have

$$P_n \subseteq P_{n+1} \quad N \cap P_n = N \cap P_{n+1} \quad N + P_n = N + P_{n+1}$$

Hence $P_n = P_{n+1}$ for large enough $n$. □

**Corollary 1.0.8.** If $M_1, \dots, M_n$ are Noetherian (resp. Artinian) $R$-modules, then $M_1 \oplus \cdots \oplus M_n$ is Noetherian (resp. Artinian).

*Proof.* By the lemma and induction. □

Recall a module homomorphism

$$\varphi : M_1 \oplus \cdots \oplus M_n \to L$$

is the same as a collection of module homomorphism $\varphi_i : M_i \to L$. This is also true for infinite direct sums (but not products!).

**Proposition 1.0.9.** For a Noetherian (resp. Artinian) ring $R$, every finitely generated $R$-module is Noetherian (resp. Artinian).

*Proof.* $M$ is finitely generated if and only if there exists a surjection $R^n \twoheadrightarrow M$ for some $n \in \mathbb{N}$. The fact that $R^n$ is Noetherian (resp. Artinian) implies that $M$ is Noetherian (resp. Artinian), as quotients of Noetherian (resp. Artinian) modules are Noetherian (resp. Artinian). □

**Definition 1.0.10** (algebra)

An *$R$-algebra* $A$ is a ring $A$ with a fixed ring homomorphism $\rho : R \to A$. We will write $r \cdot a := \rho(r)a$.

**Definition 1.0.11** (noetherian algebra)

An $R$-algebra is *Noetherian* if it is Noetherian as a ring.

**Remark 1.0.12.** Every $R$-algebra is an $R$-module.

**Example 1.0.13**

The polynomial ring $k[T_1, \dots, T_n]$ is a $k$-algebra. Do note however that it is a finitely generated by $T_1, \dots, T_n$ as a $k$-algebra, but it is infinite dimensional as a $k$-vector space.

**Definition 1.0.14** (algebra homomorphism)
$\varphi : A \to B$ is an *$R$-algebra homomorphism* if $\varphi$ is a ring homomorphism and $\varphi(r \cdot 1_A) = r \cdot 1_B$.

**Definition 1.0.15** (finitely generated algebra)
An $R$-algebra $A$ is *finitely generated* if there exists a surjective $R$-algebra homomorphism $R[T_1, \dots, T_n] \twoheadrightarrow A$ for some $n \in \mathbb{N}$.

Lecture 2

**Theorem 1.0.16** (Hilbert basis theorem)**.** Every finitely generated algebra $A$ over a Noetherian ring $R$ is Noetherian (as a ring).

For example, if $k$ is a field, then $k[T_1, \dots, T_n]$ is Noetherian.

*Proof.* It suffices to prove for $A = R[T_1, \dots, T_n]$, since every finitely generated algebra is a quotient of $R[T_1, \dots, T_n]$. Moreover, by induction, suffices to prove the result for $A = R[T]$.

Let $\mathfrak{a}$ be an ideal of $A = R[T]$. For every $i \geq 0$, define

$$\mathfrak{a}(i) = \left\{ c_0 \mid c_0 t^i + \cdots + c_i t^0 \in \mathfrak{a} \right\}$$

for the set of all leading coeffients of elements of degree $i$ in $\mathfrak{a}$ (and containing 0). In this case, $\mathfrak{a}(i) \subseteq R$ is an ideal, and we have an ascending chain of ideals

$$\mathfrak{a}(i) \subseteq \mathfrak{a}(i+1) \subseteq \cdots$$

Since $R$ is Noetherian, each $\mathfrak{a}$ is finitely generated (as an ideal), and the ascending sequence of ideal stabilises. That is,

$$\mathfrak{a}(m') = \mathfrak{a}(m)$$

for all $m' \geq m$. We write $\mathfrak{a}(i) = \langle b_{i,1}, \dots, b_{i,m_i} \rangle$, where $b_{i,j} \in R$. Let $f_{i,j} \in \mathfrak{a}$ be a polynomial of degree $i$, with leading coefficient $b_{i,j}$. Define the new ideal

$$\mathfrak{b} = \left\langle f_{i,j} \mid i \leq m, 1 \leq j \leq m_i \right\rangle \trianglelefteq R[T]$$

In this case, $\mathfrak{b}(i) = \mathfrak{a}(i)$ for all $i$. By construction, $\mathfrak{b} \subseteq \mathfrak{a}$.

Suppose for contradiction that $\mathfrak{a} \nsubseteq \mathfrak{b}$. Take $f \in \mathfrak{a} \setminus \mathfrak{b}$ of minimal degree $i$. But $\mathfrak{b}(i) = \mathfrak{a}(i)$, and so there exists $g \in \mathfrak{b}$, of degree $i$, and with the same leading coefficient as $f$. That is, $\deg(f - g) < i$. By minimality, $f - g \in \mathfrak{b}$, and so $f = (f - g) + g \in \mathfrak{b}$. Contradiction. □

Therefore, if we have a subset $S \subseteq R[T_1, \dots, T_n]/I$, then $\langle S \rangle = \langle S_0 \rangle$, where $S_0 \subseteq S$ is finite.

## 2 Tensor products

Let $M, N$ be $R$-modules. An informal definition of their tensor product is

$$M \otimes_R N = \left\{ \sum_{i=1}^{\ell} m_i \otimes n_i \;\middle|\; m_i \in M, n_i \in N \right\}$$

where we have the relations $(m_1 + m_2) \otimes n = m_1 \otimes n + m_2 \otimes n$, $m \otimes (n_1 + n_2) = m \otimes n_1 + m \otimes n_2$, and that for $r \in R$, $(rm) \otimes n = r(m \otimes n) = m \otimes (rn)$.

For example, consider $\mathbb{Z}/2 \otimes_{\mathbb{Z}} \mathbb{Z}/3$. Then

$$x \otimes y = (3x) \otimes y = x \otimes (3y) = x \otimes 0 = 0$$

and so, $\mathbb{Z}/2 \otimes_{\mathbb{Z}} \mathbb{Z}/3 = 0$. On the other hand, if we have vector spaces, then

$$\mathbb{R}^m \otimes_{\mathbb{R}} \mathbb{R}^\ell \cong \mathbb{R}^{m\ell}$$

Recall $f : M \times N \to L$ is *$R$-bilinear* if $n \mapsto f(m_0, n)$ and $m \mapsto f(m, n_0)$ are $R$-linear for all $m_0 \in M, n_0 \in N$.

**Definition 2.0.1** (tensor product of modules)
Let $M, N$ be $R$-modules, let

$$\mathcal{F} = R^{\oplus(M\times N)} = \operatorname{span}_R \left\{ e_{(m,n)} \mid m \in m, n \in N \right\}$$

be the free module indexed by $m \times n$, and define $\mathcal{K} \subseteq \mathcal{F}$ for the submodule generated by the relations (where we write $(m, n)$ for $e_{(m,n)}$)

$$\begin{aligned}(m, n_1) + (m, n_2) &= (m, n_1 + n_2)\\ (m_1, n) + (m_2, n) &= (m_1 + m_2, n)\\ r(m, n) &= (rm, n)\\ r(m, n) &= (m, rn)\end{aligned}$$

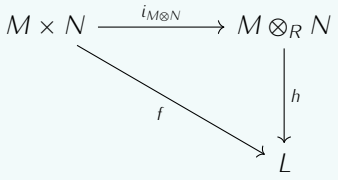
The *tensor product* is

$$M \otimes_R N := \frac{\mathcal{F}}{\mathcal{K}}$$

We have an $R$-bilinear map

$$\begin{aligned} i_{M\otimes N} : M \times N &\to M \otimes_R N\\ (m, n) &\mapsto m \otimes n\end{aligned}$$

**Proposition 2.0.2** (universal property of tensor product)**.** For every $R$-module $L$ and any $R$-bilinear map $f : M \times N \to L$, there exists a unique $R$-linear $h : M \otimes N \to L$, making the diagram




commute.

*Proof.* Uniqueness is clear, since we must have that

$$h(m \otimes n) = f(m, n)$$

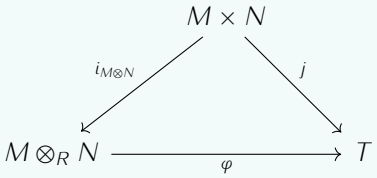
since the pure tensors generate, $h$ must be unique, if it exists. Therefore, suffices to show the above extends to an $R$-linear map $M \otimes_R N \to L$. This follows from the map

$$\begin{aligned} R^{\oplus(M\times N)} &\to L\\ e_{(m,n)} &\mapsto f(m, n)\end{aligned}$$

extending to a linear map (by the universal property of the direct sum), and that this map vanishes on $\mathcal{K}$. Therefore, $h$ extends to $M \otimes_R N$ from the pure tensors. □

**Proposition 2.0.3.** Let $M, N$ be $R$-modules, $T$ an $R$-module, $j : M \times N \to T$ an $R$-bilinear map, $(T, j)$ satisfying the universal property of tensors. Then there exists a unique $R$-linear isomorphism

$\varphi : M \otimes N \to T$, such that




commutes.

*Proof.* By the universal property of tensor product, such a map $\varphi$ exists, with $\varphi(m \otimes n) = j(m, n)$. Similarly, we have a homomorphism $\psi : T \to M \otimes_R N$. In particular,

$$\psi \circ \varphi \circ i_{M\otimes N} = i_{M\otimes N} = \mathrm{id}_{M\otimes N} \circ i_{M\otimes N}$$

In particular, by uniqueness in the universal property, we must have that $\psi \circ \varphi = \mathrm{id}_{M\otimes N}$. □

Lecture 3

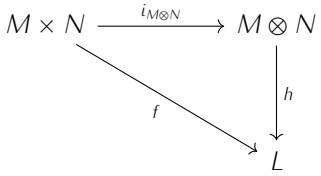
**Proposition 2.0.4.** Suppose $M, N$ are $R$-modules, then

$$\sum_i m_i \otimes n_i = 0 \in M \otimes_R N$$

if and only if for all $R$-modules $L$, and every $R$-bilinear map $f : M \times N \to L$ has

$$\sum_i f(m_i, n_i) = 0$$

*Proof.* Suppose $\sum m_i \otimes n_i = 0$, let $f : M \times N \to L$ be bilinear. Then $f$ factors through $M \times N \to M \otimes_R N$, and we can write




In this case, we have that

$$\sum_i f(m_i, n_i) = \sum_i h(i(m, n)) = \sum_i h(m_i \otimes n_i) = h\left(\sum_i m_i \otimes n_i\right) = h(0) = 0$$

Conversly, if

$$\sum_i m_i \otimes n_i \neq 0$$

then by definition,

$$\sum_i i_{m\otimes n}(m_i, n_i) \neq 0$$

□

**Example 2.0.5**

Let $k$ be a field, and consider the tensor product

$$k^m \otimes k^\ell$$

Suppose $k^m$ has basis $\{e_1, \ldots, e_m\}$, and $k^\ell$ has basis $\{f_1, \ldots, f_\ell\}$, then

$$k^m \otimes k^\ell = \mathrm{span}_k\{v \otimes w \mid v \in k^m, w \in k^\ell\} = \mathrm{span}_k\left\{e_i \otimes f_j\right\}$$

**Claim 2.0.6.** $\{e_i \otimes f_j\}$ is a basis.

*Proof.* Suppose we have

$$\sum_{ij} \alpha_{ij}(e_i \otimes f_j) = 0$$

For every $1 \leq a \leq m, 1 \leq b \leq \ell$, define a bilinear map

$$T_{ab} : k^m \times k^\ell \to k$$
$$T_{ab}((v_i), (w_j)) = v_a w_b$$

This is a $k$-bilinear map. By proposition 2.0.4,

$$0 = \sum_{i,j} \alpha_{ij} T_{ab}(e_i, f_j) = \sum_{i,j} \alpha_{ij} \delta_{ia} \delta_{jb} = \alpha_{ab}$$

□

**Example 2.0.7**

More concretely, let us consider $\mathbb{R}^2 \otimes \mathbb{R}^2$. We have a basis of size 4, given by

$$e_1 \otimes f_1, e_1 \otimes f_2, e_2 \otimes f_1, e_2 \otimes f_2$$

What do pure tensors look like?

$$(\alpha e_1 + \beta e_2) \otimes (\gamma f_1 + \delta f_2) = \alpha\gamma(e_1 \otimes f_1) + \alpha\delta(e_1 \otimes f_2) + \beta\gamma(e_2 \otimes f_1) + \beta\delta(e_2 \otimes f_2)$$

These are not generic elements of $\mathbb{R}^2 \otimes \mathbb{R}^2$, since the vectors

$$(\alpha\gamma, \alpha\delta) \quad \text{and} \quad (\beta\gamma, \beta\delta)$$

are linearly dependent. In particular,

$$e_1 \otimes f_1 + 2e_1 \otimes f_2 + 3e_2 \otimes f_1 + 4e_2 \otimes f_2$$

is *not* a pure tensor.

**Example 2.0.8** (warning)

First consider

$$\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/2$$

In this case,

$$2 \otimes 1 = 1 \otimes 2 = 1 \otimes 0 = 0$$

Now consider

$$2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/2$$

But in this case,

$$2 \otimes 1 \neq 0$$

since we can define a bilinear map

$$B : 2\mathbb{Z} \times \mathbb{Z}/2 \to \mathbb{Z}/2$$
$$B(2m, x) = mx$$

In this case,

$$B(2, 1) = 1 \cdot 1 = 1 \neq 0$$

However, if $M' \leq M, N' \leq N$ are submodules, and

$$\sum_i m_i \otimes n_i = 0$$

in $M' \otimes N'$, then

$$\sum_i m_i \otimes n_i = 0$$

in $M \otimes N$.

**Proposition 2.0.9.** If

$$\sum m_i \otimes n_i = 0 \in M \otimes_R N$$

then there are finitely generated $R$-submodules $M' \leq M, N' \leq N$, such that

$$\sum m_i \otimes n_i = 0 \in M' \otimes_R N'$$

Intuitively, a proof that the sum is zero is finite, and so it can only involve finitely many expressions. We can take them to be the generators.

*Proof.*

$$\sum m_i \otimes n_i 0 = \in M \otimes N = \frac{R^{\oplus(M \times N)}}{\mathcal{K}}$$

then

$$\sum_i e_{(m_i, n_i)} = 0 \in \mathcal{K}$$

This means that we can write the left hand side as a finite sum of the generators of $\mathcal{K}$. Taking all the elements of $M$ and $N$ which appear, gives the result. □

**Corollary 2.0.10.** Let $A, B$ be torsion-free abelian groups, then $A \otimes_{\mathbb{Z}} B$ is torsion free.

*Proof.* Suppose

$$n \cdot \left( \sum_i a_i \otimes b_i \right) = 0 \in A \otimes B$$

for some $n \geq 1$. By proposition 2.0.9, there exists finitely generated subgroups $A' \leq A, B' \leq B$, such that

$$n \cdot \left( \sum_i a_i \otimes b_i \right) = 0 \in A' \otimes B'$$

By the structure theorem of finitely generated abelian groups, $A' \cong \mathbb{Z}^r, B' \cong \mathbb{Z}^s$, and so we have that

$$A' \otimes B' \cong \mathbb{Z}^{rs}$$

which is torsion free. Contradiction. □

**Example 2.0.11**

$$\mathbb{C}^2 \otimes_{\mathbb{C}} \mathbb{C}^3 \cong \mathbb{C}^6$$

as $\mathbb{C}$-vector spaces, and we also have that $\mathbb{C}^6 \cong \mathbb{R}^1 2$ as $\mathbb{R}$-vector spaces. On the other hand,

$$\mathbb{C}^2 \otimes_{\mathbb{R}} \mathbb{C}^3 \cong \mathbb{R}^4 \otimes_{\mathbb{R}} \mathbb{R}^6 \cong \mathbb{R}^{24}$$

**Proposition 2.0.12.** 1. $M \otimes N \cong N \otimes N$

2. $(M \otimes N) \otimes P \cong M \otimes (N \otimes P) \cong M \otimes N \otimes P$, where we define $M \otimes N \otimes P$ using trilinear maps.
3. $\left(\bigoplus_i M_i\right) \otimes P \cong \bigoplus_i (M_i \otimes P)$
4. $R \otimes_R M \cong M$,

*Proof.* See examples sheet 1. □

**Example 2.0.13**

Using proposition 2.0.12, we can compute

$$\begin{aligned} R^m \otimes R^\ell &\cong \left(\oplus_{i=1}^m R\right) \otimes \left(\oplus_{j=1}^\ell R\right) \\ &\cong \oplus_{i,j} R \\ &\cong R^{m\ell} \end{aligned}$$

## 2.1 Tensor product of $R$-linear maps

**Proposition 2.1.1.** For $R$-linear maps $f : M \to M'$, $g : N \to N'$, then there exists a unique $R$-linear map

$$f \otimes g : M \otimes N \to M' \otimes N'$$

with

$$(f \otimes g)(m \otimes n) = f(m) \otimes g(n)$$

*Proof.* Uniqueness is clear since the pure tensors generate. For existence, we can use the universal property on the $R$-bilinear map

$$\begin{aligned} T : M \times N &\to M' \otimes N' \\ T(m, n) &= f(m) \otimes g(n) \end{aligned}$$

□

Lecture 4

Exercise: $(f \otimes g) \circ (h \otimes i) = (f \circ h) \otimes (h \circ i)$.

**Example 2.1.2**

Let $T : k^a \to k^c$ and $S : k^b \to k^d$ be linear. Then

$$(T \otimes S)(e_i \otimes e_j) = T(e_i) \otimes S(e_j) = \sum_{\ell,t} [T]_{\ell i}[S]_{tj}(f_\ell \otimes f_t)$$

where $[T]$ is the matrix representation of $T$. If we order the basis of $k^a \otimes k^b$ by

$$e_1 \otimes e_1, \ldots, e_1 \otimes e_c, e_2 \otimes e_1, \ldots, e_2 \otimes e_c, \ldots, e_a \otimes e_c$$

and a similar ordering for the range, then

$$[T \otimes S] = \begin{pmatrix} [T]_{11}S & \cdots & [T]_{1a}S \\ \vdots & \ddots & \vdots \\ [T]_{c1}S & \cdots & [T]_{ca}S \end{pmatrix}$$

is the *Kronecker product* of $[T]$ and $[S]$.

**Proposition 2.1.3.** Let $f : M \to M', g : N \to N'$ be $\mathbb{R}$-linear.

(i) If $f, g$ are isomorphisms, then so is $f \otimes g$,

(ii) if $f$ and $g$ are surjective, so is $f \otimes g$.

*Proof.* For (i), $(f^{-1} \otimes g^{-1}) = (f \otimes g)^{-1}$, since we have that $(f \otimes g) \circ (h \otimes i) = (f \circ h) \otimes (h \circ i)$.
For (ii), notice that $\mathrm{im}(f \otimes g)$ contains all pure tensors in $M' \otimes N'$. □

**Example 2.1.4**

If $f : \mathbb{Z} \to \mathbb{Z}$, $f(n) = pn$, then we have

$$(f \otimes \mathrm{id}) : \mathbb{Z} \otimes \mathbb{Z}/p \to \mathbb{Z} \otimes \mathbb{Z}/p$$

is the zero map, as

$$(f \otimes \mathrm{id})(a \otimes b) = (pa) \otimes b = a \otimes (pb) = a \otimes 0 = 0$$

But $\mathbb{Z} \otimes \mathbb{Z}/p \cong \mathbb{Z}/p$ which is nonzero.

## 2.2 Tensor product of algebras

Let $B, C$ be $R$-algebras. Then we have $B \otimes_R C$ as an $R$-module. We would like to define the multiplication by

$$(b \otimes c)(b' \otimes c') = (bb') \otimes (cc')$$

This is well-defined. Fix $(b, c) \in B \times \mathbb{C}$, then we have a bilinear map

$$\begin{aligned} B \times C &\to B \otimes C \\ (b', c') &\mapsto (bb') \otimes (cc') \end{aligned}$$

which gives us a map $B \otimes C \to B' \otimes C'$, with

$$b' \otimes c' \mapsto (bb') \otimes (cc')$$

It is easy to show that this then satisfies the ring axioms. Hence $B \otimes C$ is a ring. The $R$-algebra structure will be given by

$$\begin{aligned} R &\to B \otimes C \\ r &\mapsto (r1_B) \otimes 1_C = r(1_B \otimes 1_C) = 1_B \otimes (r1_C) \end{aligned}$$

**Example 2.2.1**

There is an isomorphism

$$\varphi : R[x_1, \ldots, x_n] \otimes_R R[t_1, \ldots, t_r] \cong R[x_1, \ldots, x_n, t_1, \ldots, t_r]$$

*Proof.* We have an $R$-basis for the left hand side, which is

$$x^k \otimes t^\ell$$

and we also have a $R$-basis for the right hand side,

$$x^k t^\ell$$

Define

$$\varphi(x^k \otimes t^\ell) = x^k t^\ell$$

which gives us a $R$-module isomorphism. Moreover,

$$\varphi(r \otimes 1) = r1 = 1$$

and by distributivity, suffices to show

$$\varphi((x^k \otimes t^\ell)(x^m \otimes t^n)) = x^k t^\ell x^m t^n$$

which is clear by definition. □

More generally,

$$\frac{R[x_1, \dots, x_n]}{I} \otimes \frac{R[t_1, \dots, t_r]}{J} \cong \frac{R[x_1, \dots, x_n, t_1, \dots, t_r]}{L} \cong \frac{R[x_1, \dots, x_n, t_1, \dots, t_r]}{I^e + J^e}$$

where $I^e = \langle I \rangle \trianglelefteq R[x_1, \dots, x_n, t_1, \dots, t_r]$ denotes the extension of $I$.

**Example 2.2.2**

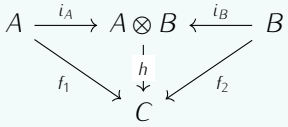
$\frac{\mathbb{C}[x,y,z]}{\langle f,g \rangle} \otimes \frac{\mathbb{C}[w,u]}{h}$ is isomorphic as $\mathbb{C}$-algebras to

$$\frac{\mathbb{C}[x, y, z, w, u]}{\langle f, g, h \rangle}$$

**Proposition 2.2.3** (universal property of tensor product of algebras). Let $A, B$ be $R$-algebras, for every $R$-algebra $C$, and $R$-algebra homomorphisms $f_1 : A \to C$ and $f_2 : B \to C$, there exists a unique $R$-algebra map

$$h : A \otimes B \to C$$

such that




commutes, where $i_A(a) = a \otimes 1$, $i_B(b) = 1 \otimes b$. Moreover, this characterises $(A \otimes B, i_A, i_B)$ uniquely (up to isomorphism).

*Proof.* $A \otimes B$ is generated, as an $R$-algebra, by

$$\{a \otimes 1 \mid a \in A\} \cup \{1 \otimes b \mid b \in B\}$$

This then implies the uniqueness of $h$, as it defines $h$ on the generators. For the existence, define the bilinear map $A \times B \to C$, given by

$$f(a, b) = f_1(a) f_2(b)$$

Using the universal property of tensor product of modules, there exists $h : A \otimes B \to C$ which is $R$-linear, with

$$h(a \otimes b) = f_1(a) f_2(b)$$

It is then easy to show that $h$ is an algebra homomorphism. □

Consider $R[x_1, \dots, x_n, t_1, \dots, t_r]$ from above. We have natural embeddings from $R[x_1, \dots, x_n]$ and $R[t_1, \dots, t_n]$. Given $f_1, f_2$ as above, we see that the image of the $x_i$ is determined by $f_1$, and the image of $t_i$ is determined by $f_2$. Therefore,

$$R[x_1, \dots, x_n, t_1, \dots, t_r] \cong \mathbb{R}[x_1, \dots, x_n] \otimes R[t_1, \dots, t_r]$$

as it satisfies the universal property.

Lecture 5

If we have $f : A \to A', g : B \to B'$ which are algebra homomorphisms, then the tensor product of $R$-linear maps,

$$f \otimes g : A \otimes B \to A' \otimes B'$$

is an $R$-algebra homomorphism. Moreover, we have $R$-algebra isomorphisms

- $(R/I) \otimes (R/J) \cong R/(I+J)$
- $A \otimes B \cong B \otimes A$,
- $(A \otimes B) \otimes C \cong A \otimes (B \otimes C)$,
- $A \otimes B^n \cong (A \otimes B)^n$,

## 2.3 Restriction and extension of scalars

### Restriction of scalars

We will have a ring homomorphisms $f : R \to S$, let $M$ be an $S$-module, so $M$ is also an $R$-module,

$$r \cdot m := f(r)m$$

for $r \in R, m \in M$. The fact that this is a module is clear by our definition, since it is just the composition

$$R \xrightarrow{f} S \longrightarrow \mathrm{End}(M)$$

> **Example 2.3.1**
> If we consider the embedding $\mathbb{R} \hookrightarrow \mathbb{C}$, then $\mathbb{C}^n$ is a $C$-vector space, but also an $\mathbb{R}$-vector space, of dimension $2n$.

### Extension of scalars

Let $f : R \to S$ be a ring homomorphism, $M$ be an $S$-module (thus an $R$-module by restriction of scalars), $N$ is an $R$-module. From this, we can form

$$M \otimes_R N$$

which is an $R$-module. In fact, $M \otimes_R N$ is also an $S$-module, with

$$s \cdot (m \otimes n) := (sm) \otimes n$$

Is this well defined? We have an $R$-bilinear map

$$\begin{aligned} M \times N &\to M \otimes_R N \\ (m, n) &\mapsto (sm) \otimes n \end{aligned}$$

By the universal property, we have a map

$$h_s : M \otimes_R N \to M \otimes_R N$$

which is $R$-linear, and $h_s(m \otimes n) = (sm) \otimes n$. Now define

$$\begin{aligned} \varphi : S &\to \mathrm{End}(M \otimes_R N) \\ \varphi(s) &= h_s \end{aligned}$$

Which is a ring homomorphism, and so, we have an $S$-module structure on $M \otimes_R N$.

> **Example 2.3.2**
> We know from before that $S \otimes_R R \cong S$ as $R$-module, with
>
> $$s \otimes r \mapsto s \cdot f(r)$$
>
> But in fact, this is also $S$-linear, since
>
> $$s' \cdot (s \otimes r) = (s's) \otimes r \mapsto s's \cdot f(r)$$
>
> For example, this implies that
>
> $$\mathbb{C} \otimes_{\mathbb{R}} \mathbb{R} \cong \mathbb{C}$$

as $\mathbb{C}$-vector spaces.

**Example 2.3.3**

If $M$ is an $S$-module, $N_i$ are $R$-modules, then

$$M \otimes_R \left( \bigoplus_i N_i \right) \cong \bigoplus_i (M \otimes_R N_i)$$

as $S$-modules.

In this case,

$$\mathbb{C} \otimes_{\mathbb{R}} \mathbb{R}^n \cong \mathbb{C}^n$$

as $\mathbb{C}$-vector spaces.

**Example 2.3.4**

Consider $\mathbb{C}^n$ as a $\mathbb{C}$-module. Restricting to $\mathbb{R}$,

$$\mathbb{C}^n \cong \mathbb{R}^{2n}$$

as $\mathbb{R}$-vector spaces. Now extending scalars,

$$\mathbb{C} \otimes_{\mathbb{R}} \mathbb{R}^{2n} \cong \mathbb{C}^{2n}$$

as $\mathbb{C}$-vector spaces.

**Example 2.3.5**

Now consider $\mathbb{R}^n$ as an $\mathbb{R}$-vector space. Extending scalars,

$$\mathbb{R}^n \otimes_{\mathbb{R}} \mathbb{C} \cong \mathbb{C}^n$$

over $\mathbb{C}$. Restricting to $\mathbb{R}$,

$$\mathbb{C}^n \cong \mathbb{R}^{2n}$$

**Example 2.3.6**

Consider $\mathbb{Z}^n$ as an $\mathbb{Z}$-module, and let $f : \mathbb{Z} \to \mathbb{Z}/2$ be the quotient map. Extending scalars,

$$(\mathbb{Z}/2) \otimes_{\mathbb{Z}} \mathbb{Z}^n \cong (\mathbb{Z}/2)^n$$

**Example 2.3.7**

Consider

$$\mathbb{C}^n \otimes_{\mathbb{R}} \mathbb{R}^\ell$$

One way to compute this:

$$\mathbb{C}^n \otimes_{\mathbb{R}} \mathbb{R}^\ell \cong_{\mathbb{R}} \mathbb{R}^{2n} \otimes \mathbb{R}^\ell \cong_{\mathbb{R}} \mathbb{R}^{2n\ell} \cong_{\mathbb{R}} \mathbb{C}^{n\ell}$$

where $\cong_{\mathbb{R}}$ denotes isomorphism as $\mathbb{R}$-vector spaces. Another way to do this:

$$\mathbb{C}^n \otimes_{\mathbb{R}} \mathbb{R}^\ell \cong_{\mathbb{C}} \mathbb{C}^n \otimes_{\mathbb{C}} \left( \mathbb{C} \otimes_{\mathbb{R}} \mathbb{R}^\ell \right) \cong_{\mathbb{C}} \mathbb{C}^n \otimes \mathbb{C}^\ell \cong_{\mathbb{C}} \mathbb{C}^{n\ell}$$

The first isomorphism is given by

$$v \otimes u \mapsto v \otimes (1 \otimes u)$$

Combining these, the isomorphism $\mathbb{C}^n \otimes_{\mathbb{R}} \mathbb{R}^\ell \to \mathbb{C}^n \otimes \mathbb{C}^\ell$ sends

$$v \otimes u \mapsto v \otimes u$$

where we use the inclusion $\mathbb{R}^\ell \hookrightarrow \mathbb{C}^\ell$.

**Proposition 2.3.8.** Let $M$ be an $S$-module, $N$ be an $R$-module, then

$$M \otimes_R N \cong M \otimes_S (S \otimes_R N)$$

as $S$-modules. In particular, the isomorphism is given by

$$\begin{aligned} m \otimes n &\mapsto m \otimes (1 \otimes n) \\ (sm) \otimes n &\mapsfrom m \otimes (s \otimes n) \end{aligned}$$

**Proposition 2.3.9.** Let $M, M'$ be $S$-modules, $N, N'$ be $R$-modules, then we have $S$-module isomorphisms

(i) $M \otimes_R N \cong N \otimes_R M$, via $m \otimes n \to n \otimes m$

(ii) $(M \otimes_R N) \otimes_R N' \cong M \otimes_R (N \otimes_R N')$

(iii) $(M \otimes_R N) \otimes_S M' \cong M \otimes_S (N \otimes_R M')$

(iv) $M \otimes_R (\bigoplus_i N_i) \cong \bigoplus_i (M \otimes_R N_i)$

*Proof.* We will prove (iii). Using proposition 2.3.8, we have

$$\begin{aligned} (M \otimes_R N) \otimes_S M' &\cong (M \otimes_S (N \otimes_R S)) \otimes_S M' \\ &\cong M \otimes_S \left( (N \otimes_R S) \otimes_S M' \right) \\ &\cong M \otimes_S (N \otimes_R M') \end{aligned}$$

□

**Example 2.3.10**

As $\mathbb{C}$-vector spaces,

$$\mathbb{C} \otimes_{\mathbb{R}} (\mathbb{R}^\ell \otimes_{\mathbb{R}} \mathbb{R}^k) \cong (\mathbb{C} \otimes_{\mathbb{R}} \mathbb{R}^\ell) \otimes_{\mathbb{C}} (\mathbb{C} \otimes_{\mathbb{R}} \mathbb{R}^k) \cong \mathbb{C}^\ell \otimes \mathbb{C}^k \cong \mathbb{C}^{\ell k}$$

**Corollary 2.3.11.** If $N, N'$ are $R$-modules, then

$$S \otimes_R (N \otimes N') \cong_S (S \otimes_R N) \otimes_S (S \otimes N')$$

*Proof.* By proposition 2.3.8 and proposition 2.3.9 (ii):

$$S \otimes_R (N \otimes_{\mathbb{R}} N') \cong (S \otimes_R N) \otimes_R N' \cong (S \otimes_R N) \otimes_S (S \otimes_R N')$$

□

Lecture 6

By induction, we have that

$$S \otimes_R (N_1 \otimes_R \cdots \otimes_R N_\ell) \cong (S \otimes_R N_1) \otimes_S \cdots \otimes_S (S \otimes_R N_1)$$

### Extension of scalars for morphisms

Let $f : N \to N'$ be $R$-linear, where $N, N'$ are $R$-modules, $M$ is an $S$-module. Then we have a map

$$\mathrm{id} \otimes f : M \otimes_R N \to M \otimes_R N'$$

In particular, it is $S$-linear, as

$$(\mathrm{id} \otimes f)(s(m \otimes n)) = (\mathrm{id} \otimes f)((sm) \otimes n) = (sm) \otimes f(n) = s(m \otimes f(n)) = s(\mathrm{id} \otimes f)(m \otimes n)$$

Given $T : \mathbb{R}^n \to \mathbb{R}^\ell$ which is an $\mathbb{R}$-linear map, $\mathbb{R}^n$ with basis $e_1, \dots, e_n$ and $\mathbb{R}^\ell$ with basis $f_1, \dots, f_\ell$. In this case, consider

$$\mathrm{id} \otimes T : \mathbb{C} \otimes \mathbb{R}^n \to \mathbb{C} \otimes \mathbb{R}^\ell$$

Note that $\mathbb{C} \otimes \mathbb{R}^n$ has basis $1 \otimes e_1, \dots, 1 \otimes e_n$. In particular,

$$(\mathrm{id} \otimes T)(1 \otimes e_i) = 1 \otimes T(e_i) = 1 \otimes \sum_{j=1}^{\ell} T_{ji} f_j = \sum_{j=1}^{\ell} T_{ji}(1 \otimes f_j)$$

Thus, $T$ and $\mathrm{id} \otimes T$ have the same matrix representation.

### Extension of scalars of algebras

Let $A, B$ be $R$-algebras. Recall that in this case, $A \otimes_R B$ is also an $R$-algebra. In fact, $A \otimes_R B$ is an $A$-algebra (and by symmetry a $B$-algebra). For example, we have

$$\begin{aligned} A &\to A \otimes_R B \\ a &\mapsto a \otimes 1 \end{aligned}$$

**Example 2.3.12**

$S \otimes_R R[x_1, \dots, x_n] \cong_S S[x_1, \dots, x_n]$ (where $\cong_S$ denotes isomorphism of $S$-algebras).

*Proof.* We already have an $S$-module isomorphism

$$\varphi : S \otimes_R R[x_1, \dots, x_n] \to S[x_1, \dots, x_n]$$

with $\varphi(s \otimes f) = sf$. It is easy to show that

$$\varphi(s \otimes 1) = s$$

and that $\varphi$ preserves multiplication. □

More generally, we have that

$$S \otimes \left( \frac{R[x_1, \dots, x_n]}{I} \right) \cong \frac{S[x_1, \dots, x_n]}{I^e}$$

where $I^e = \langle f(I) \rangle$ is the ideal generated by $I$ under the ring homomorphism $f : R \to S$.

**Proposition 2.3.13.** Suppose $A$ is an $R$-algebra, $B$ is an $S$-algebra, then $A \otimes_R B$ is an $S$-algebra. Moreover,

$$A \otimes_R B \cong_{S\text{-alg}} (A \otimes_R S) \otimes B$$

*Proof.* $A \otimes_R B$ is a $B$-algebra, and we can then restrict scalats to $S$. The isomorphism is clear from the module case, as all we need to check is it preserves multiplication. □

**Proposition 2.3.14.** Suppose $A, B$ are $R$-algebras, then

$$S \otimes_R (A \otimes_R B) \cong_{S\text{-alg}} (S \otimes_R A) \otimes_S (S \otimes_R B)$$

## 2.4 Exactness properties of the tensor product

Let $M$ be a fixed $R$-module. Define

$$\mathrm{T}_M(N) = M \otimes_R N$$

where $N$ is an $R$-module. If $f : N \to N'$ is $R$-linear, then we have an induced map

$$\mathrm{T}_M(f) = \mathrm{id}_M \otimes f : \mathrm{T}_M(N) \to \mathrm{T}_M(N')$$

Suppose we have an exact sequence

$$A \xrightarrow{f} B \xrightarrow{g} C \longrightarrow 0$$

of $R$-modules. We will show that we have an exact sequence

$$\mathrm{T}_M(A) \xrightarrow{\mathrm{T}_M(f)} \mathrm{T}_M(B) \xrightarrow{\mathrm{T}_M(g)} \mathrm{T}_M(C) \longrightarrow 0$$

That is, $\mathrm{T}_M$ is a *right exact functor* from $R$-modules to $R$-modules.

**Definition 2.4.1** (Hom)
Suppose $Q, P$ are $R$-modules, then we can define

$$\mathrm{Hom}_R(Q, P) = \{f : Q \to P \mid f \text{ is } R\text{-linear}\}$$

This is an $R$-module itself, with

$$(r \cdot \varphi)(q) = r \cdot \varphi(q)$$

**Definition 2.4.2** (Hom functors)
We have two functors,

1. $\mathrm{Hom}_R(Q, \cdot)$, where $Q$ is a fixed $R$-module,
2. $\mathrm{Hom}_R(\cdot, P)$, where $P$ is a fixed $R$-module.

Suppose we have $f : N \to N'$ which is $R$-linear, then the action on morphisms are

$$\begin{aligned} \mathrm{Hom}_R(Q, f) : \mathrm{Hom}_R(Q, N) &\to \mathrm{Hom}_R(Q, N') \\ \varphi &\mapsto f \circ \varphi =: f_*(\varphi) \end{aligned}$$

On the other hand, $\mathrm{Hom}_R(\cdot, P)$ is contravariant. That is,

$$\begin{aligned} \mathrm{Hom}_R(f, P) : \mathrm{Hom}_R(N', P) &\to \mathrm{Hom}_R(N, P) \\ \varphi &\mapsto \varphi \circ f =: f^*(\varphi) \end{aligned}$$

**Proposition 2.4.3** (left exactness of the Hom-functors). 1. If

$$0 \longrightarrow A \xrightarrow{f} B \xrightarrow{g} C$$

is exact, then so is

$$0 \longrightarrow \mathrm{Hom}_R(Q, A) \xrightarrow{\mathrm{Hom}_R(Q,f)} \mathrm{Hom}_R(Q, B) \xrightarrow{\mathrm{Hom}_R(Q,g)} \mathrm{Hom}_R(Q, C)$$

2. If
$$A \xrightarrow{f} B \xrightarrow{g} C \longrightarrow 0$$
is exact, then so is
$$0 \longrightarrow \mathrm{Hom}_R(C,P) \xrightarrow{\mathrm{Hom}_R(g,P)} \mathrm{Hom}_R(B,P) \xrightarrow{\mathrm{Hom}_R(f,P)} \mathrm{Hom}_R(A,P)$$
In both cases, we say that the respective Hom functor is *left exact*.

*Proof.* Omitted. □

**Lemma 2.4.4.** Consider a (not necessarily exact) sequence
$$A \xrightarrow{f} B \xrightarrow{g} C$$
and suppose for all $R$-module $P$, the sequence
$$\mathrm{Hom}_R(C,P) \longrightarrow \mathrm{Hom}_R(B,P) \longrightarrow \mathrm{Hom}_R(A,P)$$
is exact, then the original sequence is exact.

*Proof.* **Step 1:** let $P = C$. Then we get the sequence
$$\mathrm{Hom}_R(C,C) \longrightarrow \mathrm{Hom}_R(B,C) \longrightarrow \mathrm{Hom}_R(A,C)$$
which is exact by assumption. Under this,
$$\mathrm{id}_C \mapsto \mathrm{id}_C \circ g = g \mapsto g \circ f$$
Thus, we have that $g \circ f = 0$, and so $\mathrm{im}(f) \subseteq \ker(g)$.

**Step 2:** Let $P = \operatorname{coker} f = \frac{B}{\mathrm{im}(f)}$. In this case, we have
$$\mathrm{Hom}(C, \mathrm{coker}(f)) \longrightarrow \mathrm{Hom}(B, \mathrm{coker}(f)) \longrightarrow \mathrm{Hom}(A, \mathrm{coker}(f))$$
Let $h : B \to \mathrm{coker}(f)$ denote the quotient map. Then $h \circ f = 0$, and so by exactness, there exists $e : C \to \mathrm{coker}(f)$, with
$$\mathrm{Hom}(g, \mathrm{coker}(f))(e) = e \circ g = h$$
In particular, $\ker(g) \subseteq \ker(h) = \mathrm{im}(f)$. □

Recall that we have a bijection $\mathrm{Hom}_R(M \otimes_R N, L) \cong \mathrm{Bil}(M \times N, L)$ from the universal property of the tensor product. But
$$\mathrm{Bil}(M \times N, L) \cong \mathrm{Hom}_R(N, \mathrm{Hom}_R(M, L))$$
and so we have an isomorphism
$$\mathrm{Hom}_R(M \otimes N, L) \cong \mathrm{Hom}_R(N, \mathrm{Hom}_R(M, L))$$
sending $\varphi$ to $n \mapsto (m \mapsto \varphi(m \otimes n))$

Lecture 7

**Proposition 2.4.5.** Let $M$ be an $R$-module. Then $\mathrm{T}_M$ is a right exact functor.

*Proof.* Given an exact sequence
$$A \xrightarrow{f} B \xrightarrow{g} C \longrightarrow 0$$
Fix an $R$-module $P$. We will apply the functors $\mathrm{Hom}_R(\cdot, P)$, then the functor $\mathrm{Hom}_R(M, \cdot)$, to get the sequence
$$0 \longrightarrow \mathrm{Hom}_R(M, \mathrm{Hom}_R(C,P)) \longrightarrow \mathrm{Hom}_R(M, \mathrm{Hom}_R(B,P)) \longrightarrow \mathrm{Hom}_R(M, \mathrm{Hom}_R(A,P))$$

which is exact as the Hom functors are left exact. Using the isomorphism above, and noting that the square

$$\begin{array}{ccc} \mathrm{Hom}_R(M, \mathrm{Hom}_R(C, P)) & \longrightarrow & \mathrm{Hom}_R(M, \mathrm{Hom}_R(B, P)) \\ \downarrow & & \downarrow \\ \mathrm{Hom}_R(M \otimes C, P) & \longrightarrow & \mathrm{Hom}_R(M \otimes B, P) \end{array}$$

commutes, we have an exact sequence

$$\mathrm{Hom}_R(M \otimes C, P) \longrightarrow \mathrm{Hom}(M \otimes B, P) \longrightarrow \mathrm{Hom}(M \otimes A, P) \longrightarrow 0$$

Since $P$ is arbitrary, using lemma 2.4.4, we see that

$$\mathrm{T}_M(A) \longrightarrow \mathrm{T}_M(B) \longrightarrow \mathrm{T}_M(C) \longrightarrow 0$$

is exact, as required. □

**Remark 2.4.6.** Note on the other hand that

$$A \longrightarrow B \longrightarrow C$$

being exact does not imply that

$$\mathrm{T}_M(A) \longrightarrow \mathrm{T}_M(B) \longrightarrow \mathrm{T}_M(C)$$

is exact.

For example, consider the exact sequence

$$0 \longrightarrow \mathbb{Z} \xrightarrow{\cdot 2} \mathbb{Z}$$

This is exact, but

$$0 \longrightarrow \mathbb{Z} \otimes \mathbb{Z}/2 \xrightarrow{\cdot 2} \mathbb{Z} \otimes \mathbb{Z}/2$$

is not.

## 2.5 Flat modules - a first encounter

**Definition 2.5.1** (flat module)

An $R$-module $M$ is *flat* if for any injective $R$-module homomorphism $N \to N'$, the map $\mathrm{T}_M(f) : \mathrm{T}_M(N) \to \mathrm{T}_M(N')$ is injective.

**Example 2.5.2**

$\mathbb{Z}/2$ is not a flat $\mathbb{Z}$-module, as seen in the remark above.

**Example 2.5.3**

Free modules are flat. To see this, suppose $f : N \to N'$ is an injective $R$-linear map. Then we have the commuting square

$$\begin{array}{ccc} R^{\oplus I} \otimes N & \xrightarrow{\mathrm{id} \otimes f} & R^{\oplus I} \otimes N' \\ \updownarrow & & \updownarrow \\ (R \otimes N)^{\oplus I} & & (R \otimes N')^{\oplus I} \\ \updownarrow & & \updownarrow \\ N^{\oplus I} & \xrightarrow{f^{\oplus I}} & (N')^{\oplus I} \end{array}$$

where the vertical maps are isomorphisms, and

$$f^{\oplus I}((n_i)_{i\in I}) = (f(n_i))_{i\in I}$$

It is clear that $f^{\oplus I}$ is injective.

**Remark 2.5.4.** With this, we see that the base ring matters. $\mathbb{Z}/2$ is not a flat $\mathbb{Z}$-module, but it is a flat $\mathbb{Z}/2$-module as it is free.

**Definition 2.5.5** (torsion free)
An $R$-module is *torsion free* if for any $r \in R, m \in M$, $rm = 0$ implies that $m = 0$ or $r$ is a zero divisor.

**Proposition 2.5.6.** Flat modules are torsion free.

*Proof.* Suppose $M$ was not torsion free. Then there exists $r_0 \in R, m_0 \in M$ with $r_0$ not a zero divisor, $m_0 \neq 0$, such that $r_0 m_0 = 0$. We can define a map

$$f : R \to R$$
$$f(x) = r_0 x$$

$f$ is injective as $r_0$ is not a zero divisor. Thus, we have the square

$$\begin{array}{ccc} M \otimes R & \xrightarrow{\mathrm{id}\otimes f} & M \otimes R \\ \updownarrow & & \updownarrow \\ M & \xrightarrow[r_0\cdot]{} & M \end{array}$$

But the bottom map is not injective, as it sends $m_0$ to zero. □

For a specical case of the above:

**Proposition 2.5.7.** Let $R$ be an integral domain, $I$ a non-zero, non-unit ideal. Then $R/I$ is not flat.

*Proof.* Since $I \neq R$, $R/I$ is non-zero. Choose $x \in I \setminus 0$, and consider the map

$$f : R \to R$$
$$f(r) = xr$$

This is an injective map. But the induced map on $R \otimes (R/I) \cong R/I$ is multiplication by $x$, which is the zero map. □

**Proposition 2.5.8** (criterion for flatness)**.** Let $M$ be an $R$-module. Then teh following are equivalent:

(i) $\mathrm{T}_M$ preserves exactness of all exact sequences,

(ii) $\mathrm{T}_M$ preserves exactness of short exact sequences,

(iii) $\mathrm{T}_M$ is flat,

(iv) if $f : N \to N'$ is $R$-linear and injective, $N, N'$ are finitely generated $R$-modules, then $\mathrm{id}_M \otimes f$ is injective.

*Proof.* (i) $\implies$ (ii) $\implies$ (iii) $\implies$ (iv) is clear.

For (ii) $\implies$ (i), suppose

$$A \xrightarrow{f} B \xrightarrow{g} C$$

is exact, then we have a short exact sequence

$$0 \longrightarrow \tfrac{A}{\ker(f)} \xrightarrow{\overline{f}} B \xrightarrow{g} \operatorname{im}(g) \longrightarrow 0$$

Thus, we have a short exact sequence

$$0 \longrightarrow M \otimes \tfrac{A}{\ker(f)} \longrightarrow M \otimes B \longrightarrow M \otimes \operatorname{im}(g) \longrightarrow 0$$

That is, $\ker(\mathrm{id}_M \otimes g) = \operatorname{im}(\mathrm{id}_M \otimes \overline{f}) = \operatorname{im}(\mathrm{id}_M \otimes f)$. Thus the sequence

$$M \otimes A \longrightarrow M \otimes B \longrightarrow M \otimes C$$

is exact.

We will omit the proof of (iv) $\implies$ (iii), it can be found in the lecturer's notes.

For (iii) $\implies$ (ii), we note that this follows from $\mathrm{T}_M$ being right exact. □

**Proposition 2.5.9.** Let $f : R \to S$ be a ring homomorphism, $M$ is a flat $R$-module. Then $S \otimes_R M$ is a flat $S$-module.

*Proof.* Let $g : N \to N'$ be an injective $S$-linear map. Then the square

$$\begin{array}{ccc} (S \otimes_R M) \otimes_S N & \longrightarrow & (S \otimes_R M) \otimes_S N' \\ \updownarrow & & \updownarrow \\ M \otimes_R N & \longrightarrow & M \otimes_R N' \end{array}$$

commutes. But the bottom map is injective as $M$ is flat. □

Lecture 8

## 2.6 Further examples of tensor products

**Example 2.6.1**

First consider $x \otimes y \in \mathbb{Q} \otimes_Z (\mathbb{Z}/n)$. We can write

$$x \otimes y = n\frac{x}{n} \otimes y = \frac{x}{n} \otimes ny = 0$$

and so, $\mathbb{Q} \otimes_{\mathbb{Z}} (\mathbb{Z}/n) = 0$. We used the fact that $\mathbb{Q}$ is a *divisible group*, that is, for all $x \in \mathbb{Q}, n \in \mathbb{N}$, there exists $y \in \mathbb{Q}$ such that $ny = x$. Moreover, we also used the fact that $\mathbb{Z}/n$ is torsion.

More generally,

$$\text{divisible} \otimes \text{torsion} = 0$$

and so

$$(\mathbb{Q}/\mathbb{Z}) \otimes_Z (\mathbb{Q}/\mathbb{Z})$$

But for an $R$-module $M$ which is non-zero, if $M$ is finitely generated, then $M \otimes_R M \neq 0$.

**Example 2.6.2**

Let $V$ be a $\mathbb{Q}$ vector space, then

$$\mathbb{Q} \otimes_{\mathbb{Q}} V = V$$

But in this case, we also have that

$$\mathbb{Q} \otimes_{\mathbb{Z}} V = V$$

with $x \otimes v \mapsto xv$.

*Proof.* Every tensor in $\mathbb{Q} \otimes_{\mathbb{Z}} V$ is pure, since we can write

$$\sum \frac{a_i}{b_i} \otimes v_i = \sum \frac{1}{b_i} \otimes (a_i v_i) = \sum \frac{1}{b_i} \otimes \frac{a_i}{b_i} v_i = \sum 1 \otimes \frac{a_i}{b_i} v_i = 1 \otimes \sum \frac{a_i}{b_i}$$

Clearly this map is surjective, and it is easy to see that if $xv = 0$ then either $x = 0$ or $v = 0$. □

### Example 2.6.3

Recall that

$$M \otimes_R \left( \bigoplus_{i \in I} N_i \right) \cong \bigoplus_{i \in I} (M \otimes N_i)$$

On the other hand, if we consider the direct product, we have a map

$$\begin{aligned} M \otimes \prod_i N_i &\to \prod_i (M \otimes N_i) \\ m \otimes (n_i) &\mapsto (m \otimes n_i) \end{aligned}$$

which is in general, not an isomorphism. For example, consider

$$\mathbb{Q} \otimes_{\mathbb{Z}} \prod_{n \geq 1} \frac{\mathbb{Z}}{2^n} \to \prod_{n \geq 1} \mathbb{Q} \otimes \frac{\mathbb{Z}}{2^n}$$

But from above, $\mathbb{Q} \otimes (\mathbb{Z}/2^n) = 0$, and so the right hand side is zero. For the left hand side, take

$$g = (1, 1, \dots) \in \prod_{n \geq 1} \frac{\mathbb{Z}}{2^n}$$

Note that $g$ has infinite order, and so it generates a subgroup isomorphic to $Z$. But recall that

$$\mathbb{Q} \otimes_{\mathbb{Z}} \mathbb{Z} = \mathbb{Q}$$

With this, we have an injective map

$$\mathbb{Q} \otimes \langle g \rangle \hookrightarrow \mathbb{Q} \otimes \prod_{n \geq 1} \frac{\mathbb{Z}}{2^n}$$

We will see later that $\mathbb{Q}$ is a flat $\mathbb{Z}$-module.

### Example 2.6.4

Consider $\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C}$ as an $\mathbb{C}$-algebra, where we first restrict scalars on the right copy of $\mathbb{C}$, and extend scalars using the left copy.

Recall that as a $\mathbb{C}$-vector space,

$$\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C} = \mathbb{C} \otimes_{\mathbb{R}} \mathbb{R}^2 \cong \mathbb{C}^2$$

and we have a basis $\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C}$, which is $1 \otimes 1, 1 \otimes i$ as a $\mathbb{C}$-vector space.

To consider this as a $\mathbb{C}$-algebra, then

$$\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C} \cong \mathbb{C} \otimes_{\mathbb{R}} \frac{\mathbb{R}[t]}{\langle t^2 + 1 \rangle} \cong \frac{\mathbb{C}[t]}{\langle t^2 + 1 \rangle} = \frac{\mathbb{C}[t]}{\langle t - i \rangle \langle t + i \rangle} \cong \frac{\mathbb{C}[t]}{\langle t - i \rangle} \times \frac{\mathbb{C}[t]}{\langle t + i \rangle} \cong \mathbb{C} \times \mathbb{C}$$

where we used the Chinese remainder theorem. On a pure tensor, we have

$$(a+bi)\otimes(c+di)\mapsto(a+bi)\otimes\underbrace{[c+dt]}_{\text{coset of } c+dt}\mapsto(a+bi)[c+dt]$$

We can compute this, to get

$$P=(ac+bdit)+(ibc+tad)$$

and we then have

$$P\mapsto(ac-bd+i(bc+ad),ac+bd+i(bc-ad))$$

If we set $x=a+bi, y=c+di$, then the result is just $(xy, x\overline{y})$.

# 3 Localisation

**Definition 3.0.1** (multiplicative subset)

A *multiplicative(ly closed) subset* $S\subseteq\mathbb{R}$ such that

1. $1\in S$,
2. if $a,b\in S$, then $ab\in S$.

If $U\subseteq R$ is any set, then the *multiplicative closure* $S$ of $U$ is the set of

$$\prod_{i=1}^{n}u_i$$

where $u_i\in U$, $n\geq 0$.

**Example 3.0.2**

If $R$ is an integral domain, then $S=R\setminus\{0\}$ is multiplicative. More generally, if $\mathfrak{p}\trianglelefteq R$ is a prime ideal (of any ring $R$), then $S=R\setminus\mathfrak{p}$ is multiplicative.

**Example 3.0.3**

If $x\in R$, then $S=\{1,x,x^2,\dots\}$ is multiplicative.

**Example 3.0.4**

$\mathbb{Q}$ is obtained from $\mathbb{Z}$ by adding inverses for the elements of the multiplicative subset $\mathbb{Z}\setminus\{0\}$, and we have a ring homomorphism $\mathbb{Z}\hookrightarrow\mathbb{Q}$.

We will generalise this example to general rings $R$, and with arbitrary multiplicative subsets $S\subseteq R$. But in general, we will lose injectivity.

Lecture 9

## 3.1 Construction

**Definition 3.1.1** (localisation)

Let $S\subseteq R$ be a multiplicative set, $M$ is an $R$-module. Consider the set $M\times S$, with the relation $(m_1,s_1)\sim(m_2,s_2)$ if there exists $u\in S$, such that

$$u(s_2m_1-s_1m_2)=0$$

This is an equivalence relation, and we $S^{-1}M$ for the set of equivalence classes. We write

$$\frac{m}{s} = [(m, s)]$$

for the equivalence class. Finally, we write

$$\frac{m_1}{s_1} + \frac{m_2}{s_2} = \frac{m_1 s_2 + m_2 s_1}{s_1 s_2}$$

and

$$r \cdot \frac{m}{s} = \frac{rm}{s}$$

The above makes $S^{-1}M$ into an $R$-module. We call $S^{-1}M$ the *localisation of $M$ at $S$*.

If $M = R$, we can make $S^{-1}R$ into a ring by

$$\frac{r_1}{s_1} \cdot \frac{r_2}{s_2}$$

Next, we note that we have an $S^{-1}R$-module structure on $S^{-1}M$, via

$$\frac{r}{s} \cdot \frac{m}{t} = \frac{rm}{st}$$

We have localisation maps:

$$R \to S^{-1}R$$
$$r \mapsto \frac{r}{1}$$

which is a ring homomorphism, and

$$M \to S[-1]M$$
$$m \mapsto \frac{m}{1}$$

which is an $R$-linear map.

We check that $\sim$ above defines an equivalence relation: Reflexivity and symmetry are clear. Say $(m_1, s_1) \sim (m_2, s_2)$ and $(m_2, s_2) \sim (m_3, s_3)$. That is, there exists $u, v \in S$ such that

$$u(s_2 m_1 - s_1 m_2) = v(s_3 m_2 - s_2 m_3) = 0$$

Multiplying the first term by $vs_3$ and the second by $us_1$, we get

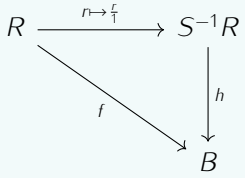
$$uvs_2 s_3 m_1 = uvs_3 s_1 m_2$$
$$uvs_1 s_3 m_2 = uvs_1 s_2 m_3$$

and so, we have that

$$uvs_2(s_3 m_1 - s_1 m_3) = 0$$

Since $S$ is multiplicatively closed, we are done.

**Proposition 3.1.2** (universal property of $S^{-1}R$)**.** Let $U \subseteq R$ be any subset, and let $S \subseteq R$ be the multiplicative closure of $U$. Let $f : R \to B$ be a ring homomorphism, such that $f(u)$ is a unit for all $u \in U$. Then there exists a unique ring homomorphism $h : S^{-1}R \to B$, such that the diagram



commutes. That is,

$$f(r) = h\left(\frac{r}{1}\right)$$

Another way of thinking about this is that we have a bijection

$$\mathrm{Hom}_{\mathrm{Ring}}(S^{-1}R, B) \leftrightarrow \{\varphi : R \to B \text{ ring hom., with } \varphi(U) \subseteq B^\times\}$$

given by sending $f$ to $r \mapsto f\left(\frac{r}{1}\right)$.

*Proof.* Let $f : R \to B$ be a ring homomorphism, with $f(U) \subseteq B^\times$. In this case, $f(S) \subseteq B^\times$ as well. We want $h : S^{-1}R \to B$, with

$$f(r) = h\left(\frac{r}{1}\right)$$

First, such $h$ must satisfy:

$$1 = h(1) = h\left(\frac{1}{s} \cdot \frac{s}{1}\right) = h\left(\frac{1}{s}\right) f(s)$$

Thus, we must have that $h(1/s) = f(s)^{-1}$. With this, we have

$$h\left(\frac{r}{s}\right) = h\left(\frac{r}{1}\right) h\left(\frac{1}{s}\right) = f(r)f(s)^{-1}$$

But we need to check if $h$ is well defined. That is, if $r_1/s_1 = r_2/s_2$, then there exists $t \in S$ such that $t(s_2r_1 - s_1r_2) = 0$, or equivalently,

$$ts_2r_1 = ts_1r_2$$

Applying $f$, we get

$$f(t)f(s_2)f(r_1) = f(t)f(s_1)f(r_2)$$

But every element in the above equality are in $B^\times$, and so we are done. It is easy to check that $h$ is a ring homomorphism. □

**Proposition 3.1.3.** If $(A, j)$ satisfies the same universal property of $(S^{-1}R, \iota)$, where $\iota(r) = r/1$, then there exists an isomorphism $S^{-1}R \to A$, sending

$$\frac{r}{s} \mapsto j(r)j(s)^{-1}$$

### Facts

1. Take $r/s \in S^{-1}R$, then
$$\frac{r}{s} = \frac{0}{1} \iff \text{there exists } u \in S \text{ with } ur = 0$$
2. $S^{-1}R = 0$ if and only if $0 \in S$.
3. $$\ker(\iota : R \to S^{-1}R) = \{r \in R \mid \text{ there exists } u \in S \text{ with } ur = 0\}$$
4. In particular, $\iota$ is injective if and only if $S$ does not contain any zero divisors.
5. $\iota$ is always an epimorphism[1], but usually not surjective. For example, $\iota : \mathbb{Z} \to \mathbb{Q}$ is an epimorphism. If we have $f, g : \mathbb{Q} \to A$ ring homomorphisms, with $f \circ \iota = g \circ \iota$, then $f = g$.

**Example 3.1.4**

For $f \in R$, let $S = \{f^n \mid n \geq 0\}$. Then we define $R_f = S^{-1}R$.

[1] A morphism $f : X \to Y$ (in some category) is called an *epimorphism* if for all $g_1, g_2 : Y \to Z$, with $g_1 \circ f = g_2 \circ f$, we have $g_1 = g_2$.

If $R = \mathbb{Z}, f = 2$, then

$$R_f = \left\{ \frac{a}{2^n} \mid a \in \mathbb{Z}, n \geq 0 \right\} = \mathbb{Z}\left[\frac{1}{2}\right]$$

**Notation 3.1.5.** In this course, we will write:

- $\mathbb{Z}/n$ for the finite ring,
- $\mathbb{Z}_2$ for the 2-adic integers,
- $\mathbb{Z}[1/2]$ for the above ring.

### Example 3.1.6

For a ring $R$, let $\operatorname{Spec}(R)$ denote its prime spectrum. For $\mathfrak{p} \in \operatorname{Spec}(R)$, we can let $S = R \setminus \mathfrak{p}$, and we write $R_{\mathfrak{p}} = (R \setminus \mathfrak{p})^{-1}R$.

If $R = \mathbb{Z}$, $p = \langle 3 \rangle$, then

$$\mathbb{Z}_{\langle 3 \rangle} = \left\{ \frac{a}{b} \mid a, b \in \mathbb{Z}, 3 \nmid b \right\}$$

**Proposition 3.1.7.** If $M$ is an $R$-module, $S \subseteq R$ a multiplicative subset, then we have an isomorphism:

$$\begin{aligned} \underset{R}{\otimes} M &\to S^{-1}M \\ \frac{r}{s} \otimes m &\mapsto \frac{rm}{s} \end{aligned}$$

*Proof.* We can define a bilinear map

$$\begin{aligned} S^{-1}R \times M &\to S^{-1}M \\ \left(\frac{r}{s}, m\right) &\mapsto \frac{rm}{s} \end{aligned}$$

and thus, by the universal property we haev $\varphi : S^{-1}R \otimes_R M \to S^{-1}M$. This is $R$-linear, and it is easy to see that $\varphi$ is also $S^{-1}R$-linear. It is clear that $\varphi$ is surjective, since

$$\varphi\left(\frac{1}{s} \otimes m\right) = \frac{m}{s}$$

We want to show that every tensor

$$t = \sum_i \frac{r_i}{s_i} \otimes m_i \in S^{-1}R \otimes_R M$$

is prime. Define $s = \prod_i s_i$, and $t_j = \prod_{i \neq j} s_i$. In this case,

$$\begin{aligned} \sum \frac{r_i}{s_i} \otimes m_i &= \sum \frac{1}{s_i} \otimes (r_i m_i) \\ &= \sum \frac{t_i}{s} \otimes (r_i m_i) \\ &= \frac{1}{s} \otimes \left( \sum_i r_i t_i m_i \right) \end{aligned}$$

Using this, if

$$\varphi\left(\frac{1}{s} \otimes m\right) = \frac{m}{s} = 0 = \frac{0}{1}$$

That is, there exists $u \in S$, such that $um = 0$. In this case,

$$\frac{1}{s} \otimes m = \frac{u}{us} \otimes m = \frac{1}{us} \otimes (um) = 0$$

□

With this, $S^{-1}R \otimes (\cdots)$ acts on $R$-modules. But in fact, it also acts on $R$-linear maps.

**Proposition 3.1.8** (localisation is a functor)**.** Let $M$ be an $R$-module, $S \subseteq R$ a multiplicative subset. Let $f : N \to N'$ be an $R$-linear map. Then the following square commutes:

$$\begin{array}{ccc} S^{-1}R \otimes N & \xrightarrow{\mathrm{id}_{S^{-1}R} \otimes f} & S^{-1}R \otimes N' \\ \downarrow \sim & & \downarrow \sim \\ S^{-1}N & \xrightarrow[S^{-1}(f)]{} & S^{-1}N' \end{array}$$

In particular,

$$(S^{-1}f)\left(\frac{n}{s}\right) = \frac{f(n)}{s}$$

With this, the functors $S^{-1}R \otimes (\cdot)$ and $S^{-1}(\cdot)$ are naturally isomorphic.

**Remark 3.1.9.** Let $A$ be an $R$-algebra, $S^{-1}R \otimes A \to S^{-1}A$ is $S^{-1}R$-linear, and also an isomorphism of $S^{-1}R$-algebras.

**Lemma 3.1.10.** If $M$ is an $S^{-1}R$-module, then, we can restrict scalars on $M$ from $S^{-1}R$ to $R$, then apply $S^{-1}(\cdot)$. Then

$$S^{-1}M \cong M$$

as $S^{-1}R$-modules. Equivalently,

$$M \cong S^{-1}R \otimes M$$

as $S^{-1}R$-modules.

*Proof.* We can see that the map

$$M \to S^{-1}M \tag{1}$$

$$m \mapsto \frac{m}{1} \tag{2}$$

is $S^{-1}R$-linear. Surjectivity and injectivity are clear. □

**Proposition 3.1.11.** Let $M$ be an $R$-module, $L$ an $S^{-1}R$-module, $f : M \to L$ is $R$-linear. Then there exists a unique $h : S^{-1}M \to L$ which is $S^{-1}R$-linear, such that

$$f(m) = h\left(\frac{m}{1}\right)$$

*Proof.* We know that $S^{-1}(\cdot) \otimes S^{-1}R \otimes (\cdot)$, and so it suffices to prove the result for the tensor product. With this, the localisation map is

$$\iota : M \to S^{-1}R \otimes M$$

$$m \mapsto \frac{1}{1} \otimes m$$

Let $f : M \to L$ be $R$-linear. We then have that

$$h : \mathrm{id}_{S^{-1}R} \otimes f : S^{-1}R \otimes_R M \to S^{-1}R \otimes_R L$$

But the previous lemma shows that $S^{-1}R \otimes_R L \cong L$ as $S^{-1}R$-modules. In particular,

$$h\left(\frac{r}{s} \otimes m\right) = \frac{r}{s} f(m)$$

For the uniqueness of $h$, it follows from the fact that elements of the form $\frac{1}{1} \otimes m$ generate $S^{-1}R \otimes_R M$ as an $S^{-1}R$-module. □

**Proposition 3.1.12** (the functor $S^{-1}R$ is exact)**.** If

$$A \xrightarrow{f} B \xrightarrow{g} C$$

is an exact sequence of $R$-modules, then

$$S^{-1}A \xrightarrow{S^{-1}f} S^{-1}B \xrightarrow{S^{-1}g} S^{-1}C$$

is an exact sequence of $S^{-1}R$-modules.

*Proof.*

$$(S^{-1}g) \circ (S^{-1}f) = S^{-1}(g \circ f) = S^{-1}(0) = 0$$

and so $\operatorname{im}(S^{-1}f) \subseteq \ker(S^{-1}g)$. Let

$$\frac{b}{s} \in \ker(S^{-1}g)$$

Then

$$\frac{g(b)}{s} = \frac{0}{1}$$

That is, there exists $u \in S$, such that $u \cdot g(b) = 0$. But $g$ is $R$-linear, $u \in R$, and so $g(ub) = 0$, which means that $ub \in \ker(g) = \operatorname{im}(f)$. Thus, there exists $a \in A$ such that $f(a) = ub$. Now

$$\frac{b}{s} = \frac{ub}{us} = \frac{f(a)}{us} = S^{-1}f\left(\frac{a}{us}\right) \in \operatorname{im}(S^{-1}f)$$

□

Equivalently, $S^{-1}R$ is a flat $R$-module. Suppose $\iota : N \to M$ is the inclusion map, then

$$S^{-1}\iota : S^{-1}N \to S^{-1}M$$

is injective, and so the expression

$$\frac{n}{s}$$

makes sense in $S^{-1}N$ *and* $S^{-1}(M)$.

**Proposition 3.1.13.** Let $M$ be an $R$-module, $N, P$ submodules of $M$. Then

(i) $S^{-1}(N + P) = S^{-1}N + S^{-1}P$.

(ii) $S^{-1}(N \cap P) = S^{-1}N \cap S^{-1}P$,

(iii) $(S^{-1}M)/(S^{-1}N) \cong S^{-1}(M/N)$ as $S^{-1}R$ modules via

$$\frac{m}{s} + S^{-1}N \leftrightarrow \frac{m + N}{s}$$

*Proof.* For (i), the left hand side consists of elements of the form $\frac{n+p}{s}$, and the right hand side consists of elements of the form $\frac{n}{s_1} + \frac{p}{s_2}$. The result is then clear.

For (ii), $\subseteq$ is clear. Given $x \in S^{-1}N \cap S^{-1}P$, that is,

$$x = \frac{n}{s_1} = \frac{p}{s_2}$$

for $n \in N, p \in P, s_1, s_2 \in S$. But then there exists $u \in S$, such that $us_2n = us_1p =: w \in N \cap P$. With this,

$$x - \frac{n}{s_1} = \frac{us_2n}{us_1s_2} = \frac{w}{us_1s_2} \in S^{-1}(N \cap P)$$

For (iii), consider the exact sequence

$$0 \longrightarrow N \longrightarrow M \longrightarrow M/N \longrightarrow 0$$

Applying the exact functor $S^{-1}$,

$$0 \longrightarrow S^{-1}N \longrightarrow S^{-1}M \longrightarrow S^{-1}(M/N) \longrightarrow 0$$

But this immediately gives that

$$S^{-1}(M/N) \cong \frac{S^{-1}M}{S^{-1}N}$$

as $S^{-1}R$-modules. Computing the respective maps gives the result. □

**Proposition 3.1.14.** If $M, N$ are $R$-modules, then

$$S^{-1}M \otimes_{S^{-1}R} S^{-1}N \cong S^{-1}(M \otimes_R N)$$

*Proof.* We have the isomorphism from extension of scalars:

$$(S^{-1}R \otimes_R M) \otimes_{S^{-1}R} (S^{-1}R \otimes_R N) \cong S^{-1}R \otimes_R (M \otimes_R N)$$

□

A special case of this is that if $\mathfrak{p}$ is a prime ideal of $R$, then

$$M_{\mathfrak{p}} \otimes_{R_{\mathfrak{p}}} N_{\mathfrak{p}} = (M \otimes_R N)_{\mathfrak{p}}$$

## 3.2 Extension and contraction of ideals

Recall if $f : A \to B$ is a ring homomorphism, we define the *contraction of* $\mathfrak{b} \trianglelefteq B$ as

$$\mathfrak{b}^c = f^{-1}(\mathfrak{b}) \trianglelefteq A$$

and the *extension of* $\mathfrak{a} \trianglelefteq A$ as

$$\mathfrak{a}^e = \langle f(\mathfrak{a}) \rangle \trianglelefteq B$$

In examples sheet 1, we have a bijection

$$\{\text{contracted ideals of } A\} \leftrightarrow \{\text{extended ideals of } B\}$$

To see this, we have that an ideal $\mathfrak{a}$ is contracted if and only if $\mathfrak{a} = \mathfrak{a}^{ec}$, and an ideal $\mathfrak{b}$ is extended if and only if $\mathfrak{b} = \mathfrak{b}^{ce}$, and so the bijection is given by extension/contraction.

Let $S$ be a multiplicative subset of $R$, and we will consider the ring homomorphism $R \to S^{-1}R$, given by $r \mapsto r/1$. For an ideal $\mathfrak{a}$ of $R$, we have the *extension*

$$\mathfrak{a}^e = S^{-1}\mathfrak{a} \trianglelefteq S^{-1}R$$

and for an ideal $\mathfrak{b}$ of $S^{-1}R$, we have the contraction $\mathfrak{b}^c \trianglelefteq R$.

**Proposition 3.2.1.**

$$\mathfrak{a}^e = S^{-1}\mathfrak{a} = \left\{ \frac{a}{s} \;\middle|\; a \in \mathfrak{a}, s \in S \right\}$$

Lecture 11

*Proof.* $\mathfrak{a}^e$ is the ideal generated by $a/1$ for $a \in \mathfrak{a}$, and so $\supseteq$ holds. But the right hand side is already an ideal, and so by minimality, equality holds. □

**Proposition 3.2.2.** $\mathfrak{a}^{ec} = \bigcup_{s\in S}(\mathfrak{a} : s)$ where $(\mathfrak{a} : s) = \{r \in R \mid rs \in \mathfrak{a}\}$.

*Proof.* Take $r \in \bigcup_{s\in S}(\mathfrak{a} : s)$. That is, $rs = a \in \mathfrak{a}$, and so in $S^{-1}R$,

$$\frac{rs}{1} = \frac{a}{1} \implies \frac{r}{1} = \frac{a}{s} \in \mathfrak{a}^e$$

and so $r \in \mathfrak{a}^{ec}$. Conversely, if $r \in \mathfrak{a}^{ec}$, then

$$\frac{r}{1} = \frac{a}{s}$$

for some $a \in \mathfrak{a}, s \in S$. But this means that there exists $u \in S$, such that $urs = ua$. With this, $r \in (\mathfrak{a} : us)$, $us \in S$ as $S$ is multiplicative. □

Now suppose $\mathfrak{b}$ is an ideal of $S^{-1}R$. Then

$$\mathfrak{b}^c = \left\{ r \in \mathbb{R} \;\middle|\; \frac{r}{1} \in \mathfrak{b} \right\}$$

**Proposition 3.2.3.** $\mathfrak{b}^{ce} = \mathfrak{b}$.

*Proof.* $\subseteq$ always holds. Take $r/s \in \mathfrak{b}$, then $r/1 \in \mathfrak{b}$. Thus, $r \in \mathfrak{b}^c$, and so $r/1 \in \mathfrak{b}^{ce}$, which means that $r/s \in \mathfrak{b}^{ce}$. □

**Proposition 3.2.4.** Consider the loclisation map $R \to S^{-1}R$, then

(i) Every ideal of $S^{-1}R$ is extended.

(ii) An ideal $\mathfrak{a}$ of $R$ is contracted if and only if the image of $S$ in $R/\mathfrak{a}$ contains no zero divisors of $R/\mathfrak{a}$.

(iii) $\mathfrak{a}^e = S^{-1}R$ if and only if $\mathfrak{a} \cap S \neq \varnothing$.

(iv) We have a bijection:

$$\begin{aligned} \{\mathfrak{p} \in \operatorname{Spec}(R) \mid \mathfrak{p} \cap S = \varnothing\} &\leftrightarrow \operatorname{Spec}(S^{-1}R) \\ \mathfrak{p} &\mapsto \mathfrak{p}^e \\ \mathfrak{q}^c &\hookleftarrow \mathfrak{q} \end{aligned}$$

*Proof.* (i) Follows from proposition 3.2.3. For (ii), $\mathfrak{a}$ is contracted if and only if $\mathfrak{a}^{ec} \subseteq \mathfrak{a}$. But

$$\mathfrak{a}^{ec} = \bigcup_{s\in S}(\mathfrak{a} : s)$$

Thus, $\mathfrak{a}^{ec} \subseteq \mathfrak{a}$ if and only if: for all $r \in R$, if $Sr \cap \mathfrak{a} \neq \varnothing$, then $r \in \mathfrak{a}$. But $Sr \cap \mathfrak{a} \neq$ is true if and only if $0 + \mathfrak{a}$ is in the image of $S$, and $r \in \mathfrak{a}$ is the same as $r + \mathfrak{a} = 0$. Thus, $\mathfrak{a}$ is contracted if and only if the image of $S$ in $R/\mathfrak{a}$ contains no zero divisors.

For (iii), suppose $\mathfrak{a} \cap S \neq \varnothing$. Choose $x \in \mathfrak{a} \cap S$, then

$$1 = \frac{x}{x} \in \mathfrak{a}^e$$

Conversely, if $\mathfrak{a}^e = S^{-1}R$. Then $1 \in \mathfrak{a}^e$, and so

$$\frac{1}{1} = \frac{a}{s}$$

for some $a \in \mathfrak{a}, s \in S$, and so there exists $u \in S$ such that $us = ua$. But $us \in S$ as it is multiplicative, $ua \in \mathfrak{a}$ as it is an ideal.

For (iv), first consider the contraction map $\mathrm{Spec}(S^{-1}R) \to \{\mathfrak{p} \in \mathrm{Spec}(R) \mid \mathfrak{p} \cap S = \varnothing\}$. This makes sense as the contraction of a prime ideal is prime, and if $\mathfrak{p} \in \mathrm{Spec}(R)$ is contracted, by (ii), we see that $S \cap \mathfrak{p}$ is empty, since $R/\mathfrak{p}$ is an integral domain, and so the only zero divisor is zero.

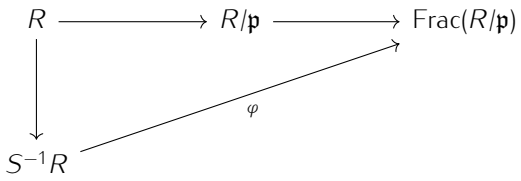
Moreover, this map is injective, since it has a left inverse, as all ideals in $S^{-1}R$ are extended ideals, and so $\mathfrak{q}^{ce} = \mathfrak{q}$. In the other direction, for a prime ideal $\mathfrak{p} \in \mathrm{Spec}(R)$, with $\mathfrak{p} \cap S = \varnothing$, we have seen that $\mathfrak{p}$ is contracted, and so $\mathfrak{p}^{ec} = \mathfrak{p}$. With this, all we need to show is that $\mathfrak{p}^e$ is prime.

We would like to show that $(S^{-1}R)/\mathfrak{p}^e$ is an integral domain. We know that $\mathfrak{p}^e$ is not all of $S^{-1}R$, and so $(S^{-1}R)/\mathfrak{p}^e$ is not the zero ring. So we need to show that $(S^{-1}R)/\mathfrak{p}^e$ has no zero divisors. We will do this by embedding $(S^{-1}R)/\mathfrak{p}^e$ into $\mathrm{Frac}(R/\mathfrak{p})$.

Now consider the composition map

$$R \longrightarrow R/\mathfrak{p} \longrightarrow \mathrm{Frac}(R/\mathfrak{p})$$

This has the property that the elements of $S$ are sent to units, since $S \cap \mathfrak{p} = \varnothing$. Using the universal property of $S^{-1}R$, we hava an induced map




In particular,

$$\varphi\left(\frac{r}{s}\right) = \frac{r+\mathfrak{p}}{s+\mathfrak{p}}$$

It suffices to show that $\ker(\varphi) = \mathfrak{p}^e$. First, we see that $\mathrm{im}(\varphi) \subseteq \overline{S}^{-1}(R/\mathfrak{p})$, where $\overline{S}$ is the image of $S$ in $S^{-1}R$. With this, we cam consider $\varphi : S^{-1}R \to \overline{S}^{-1}(R/\mathfrak{p})$. Take $r/s \in \ker(\varphi)$. That is,

$$\frac{r+\mathfrak{p}}{s+\mathfrak{p}} = \frac{0}{1} \in \overline{S}^{-1}(R/\mathfrak{p})$$

Then there exists $u + \mathfrak{p} \in \overline{S}$, such that

$$(u+\mathfrak{p})(r+\mathfrak{p}) = (ur) + \mathfrak{p} = 0$$

That is, $ur \in \mathfrak{p}$. Then we have that

$$\frac{r}{s} = \frac{ur}{us} \in \mathfrak{p}^e$$

Conversely, take $x \in \mathfrak{p}^e$. Then $x = p/s$, and

$$\varphi(x) = \frac{p+\mathfrak{p}}{s+\mathfrak{p}} = 0$$

and so $x \in \ker(\varphi)$. □

### An application

If $I \trianglelefteq R$ is an ideal, then the *radical of* $I$ is

$$\sqrt{I} = \{r \in R \mid \exists m \geq 1 \text{ such that } r^m \in I\}$$

**Proposition 3.2.5.**

$$\sqrt{I} = \bigcap_{I \leq \mathfrak{p} \in \mathrm{Spec}(R)} \mathfrak{p}$$

*Proof.* Take $x \in \sqrt{I}$, then $x^n \in I$, and so for every $\mathfrak{p} \in \operatorname{Spec}(R)$, if $I \subseteq \mathfrak{p}$, then $x^n \in \mathfrak{p}$, and so $x \in \mathfrak{p}$. That is, $\subseteq$ holds. For the other inclusion, take $x \in R$, $x \notin \sqrt{I}$. We know that $I \neq R$, and $R/I$ is not the zero ring. Let $\overline{x} \in R/I$ be the image of $x$. Consider

$$(R/I)_{\overline{x}} = \{\overline{x}^n\}^{-1}(R/I)$$

This is not the zero ring, since we did not invert zero. Therefore, $(R/I)_{\overline{x}}$ has a prime ideal, which corresponds to a prime ideal of $R/I$ which avoids $\overline{x}$, which in turn, corresponds to a prime ideal of $R$, which contains $I$, and avoids $x$. □

Lecture 12

## 3.3 Local properties

**Definition 3.3.1** (local ring)

A ring $R$ is *local* if it has a unique maximal ideal. We write $(R, \mathfrak{m})$ for the local ring $R$ with maximal ideal $\mathfrak{m}$.

**Example 3.3.2**

Let $\mathfrak{p} \in \operatorname{Spec}(R)$. Then recall that we have a bijection

$$\{\mathfrak{q} \in \operatorname{Spec}(R) \mid \mathfrak{q} \subseteq \mathfrak{p}\} \leftrightarrow \operatorname{Spec}(R_{\mathfrak{p}})$$

given by extension and contraction. With this, all prime ideals of $R_{\mathfrak{p}}$ are contained in $\mathfrak{p}R_{\mathfrak{p}}$. Thus, $(R_{\mathfrak{p}}, \mathfrak{p}R_{\mathfrak{p}})$ is a local ring.

In particular, $\mathbb{Z}_{\langle 2\rangle}$ is a local ring, and the unique maximal ideal is

$$\langle 2\rangle \mathbb{Z}_{\langle 2\rangle} = \left\{ \frac{2a}{b} \;\middle|\; a, b \in \mathbb{Z}, 2 \nmid b \right\}$$

**Proposition 3.3.3.** Let $M$ be an $R$-module. Then the following are equivalent:

(i) $M = 0$,

(ii) $M_{\mathfrak{p}} = 0$ for all $\mathfrak{p} \in \operatorname{Spec}(R)$,

(iii) $M_{\mathfrak{m}} = 0$ for all $\mathfrak{m} \in \operatorname{maxSpec}(R)$.

That is, being zero is a local property (i.e. it is localisable and local to global).

*Proof.* The implications $(i) \implies (ii) \implies (iii)$ is clear. Suppose (iii) holds, and suppose for contradiction there exists $m \in M$ non-zero. Consider

$$\operatorname{Ann}_R(m) = \{r \in R \mid rm = 0\} \trianglelefteq R$$

Since $m \neq 0$, $1 \notin \operatorname{Ann}_R(m)$. Take a maximal ideal $\mathfrak{m}$ containing $\operatorname{Ann}_R(m)$. In this case,

$$\frac{m}{1} = 0 \in M_{\mathfrak{m}}$$

That is, $um = 0$ for some $u \in R \setminus \mathfrak{m}$. But in this case, $u \notin \operatorname{Ann}_R(m)$. Contradiction. □

**Proposition 3.3.4.** Lte $f : M \to N$ be an $R$-linear map. Then the following are equivalent:

(i) $f$ is injective,

(ii) $f_{\mathfrak{p}} : M_{\mathfrak{p}} \to N_{\mathfrak{p}}$ is injective for every $\mathfrak{p} \in \operatorname{Spec}(R)$,

(iii) $f_{\mathfrak{m}} : M_{\mathfrak{m}} \to N_{\mathfrak{m}}$ is injective for every $\mathfrak{m} \in \operatorname{maxSpec}(R)$,

The same statements holds for surjectivity.

Recall

$$f_{\mathfrak{p}}\left(\frac{m}{s}\right) = \frac{f(m)}{s}$$

*Proof.* Suppose (i) holds. Since localising at $\mathfrak{p}$ is an exact functor, (ii) follows. (ii) implies (iii) is by definition.

Suppose (iii) holds. We have the exact sequence

$$0 \longrightarrow \ker(f) \hookrightarrow M \xrightarrow{f} N$$

Localising at $\mathfrak{m}$, we get

$$0 \longrightarrow \ker(f)_{\mathfrak{m}} \longrightarrow M_{\mathfrak{m}} \xrightarrow{f_{\mathfrak{m}}} N_{\mathfrak{m}} \tag{$*$}$$

which is exact as localisation is an exact functor. But $(*)$ shows that

$$\ker(f_{\mathfrak{m}}) = \ker(f)_{\mathfrak{m}}$$

But we assumed $\ker(f_{\mathfrak{m}}) = 0$, and so $\ker(f)_{\mathfrak{m}} = 0$ for all maximal ideals $\mathfrak{m}$. Thus, by proposition 3.3.3, $\ker(f) = 0$. □

**Proposition 3.3.5.** Let $M$ be an $R$-module. Then the following are equivalent:

(i) $M$ is a flat $R$-module,

(ii) $M_{\mathfrak{p}}$ is a flat $R_{\mathfrak{p}}$-module for all $\mathfrak{p} \in \operatorname{Spec}(R)$,

(iii) $M_{\mathfrak{m}}$ is a flat $R_{\mathfrak{m}}$-module for all $\mathfrak{m} \in \operatorname{maxSpec}(R)$.

*Proof.* For (i) $\implies$ (ii), since $M_{\mathfrak{p}} \cong R_{\mathfrak{p}} \otimes_R M$ as $R_{\mathfrak{p}}$-modules, and we have shown that extension of scalars preserves flatness. As usual, (ii) $\implies$ (iii) is trivial.

Suppose (iii) holds. Suppose $f : N \to P$ is $R$-linear and injective. Fix a maximal ideal $\mathfrak{m} \in \operatorname{maxSpec}(R)$. Then $f_{\mathfrak{m}} : N_{\mathfrak{m}} \to P_{\mathfrak{m}}$ is injective by proposition 3.3.4. Then

$$N_{\mathfrak{m}} \otimes M_{\mathfrak{m}} \xrightarrow{f_{\mathfrak{m}} \otimes \mathrm{id}} P_{\mathfrak{m}} \otimes M_{\mathfrak{m}}$$

is injective by (iii). But we have isomorphisms $(N \otimes_R M)_{\mathfrak{m}} \cong N_{\mathfrak{m}} \otimes_{R_{\mathfrak{m}}} M_{\mathfrak{m}}$, and using this,

$$\begin{array}{ccc} N_{\mathfrak{m}} \otimes M_{\mathfrak{m}} & \xrightarrow{f_{\mathfrak{m}} \otimes \mathrm{id}} & P_{\mathfrak{m}} \otimes M_{\mathfrak{m}} \\ \downarrow \sim & & \downarrow \sim \\ (N \otimes_R M)_{\mathfrak{m}} & \xrightarrow{(f \otimes \mathrm{id})_{\mathfrak{m}}} & (P \otimes_R M)_{\mathfrak{m}} \end{array}$$

the bottom map must be injective. But then $(f \otimes \mathrm{id})_{\mathfrak{m}}$ is injective for all $\mathfrak{m}$, and so $f \otimes \mathrm{id}$ is injective by proposition 3.3.4. □

**Example 3.3.6**

An $R$-module $M$ is *locally free* if $M_{\mathfrak{p}}$ is a free $R_{\mathfrak{p}}$ module for every $\mathfrak{p} \in \operatorname{Spec}(R)$.

Take $R = \mathbb{C} \times \mathbb{C}$. The set of prime ideals of $R$ is just

$$\{\mathbb{C} \times 0, 0 \times \mathbb{C}\}$$

But then we have a ring homomorphism

$$\begin{aligned} \mathbb{C} \times \mathbb{C} &\to \mathbb{C} \\ (a, b) &= b \end{aligned}$$

This sends $\mathbb{C}\times\mathbb{C}\setminus\mathbb{C}\times 0$ to units, and so we have a ring homomorphism

$$\begin{aligned}(\mathbb{C}\times\mathbb{C})_{\mathbb{C}\times 0} &\to \mathbb{C}\\ \frac{(a,b)}{(c,d)} &\mapsto \frac{b}{d}\end{aligned}$$

This is a bijection. With this, $(\mathbb{C}\times\mathbb{C})_{\mathbb{C}\times 0}\cong(\mathbb{C}\times\mathbb{C})_{0\times\mathbb{C}}$ are fields, and so every $\mathbb{C}\times\mathbb{C}$-module $M$ is locally free.

Now consider $M=\mathbb{C}\times\{0\}$ as an $\mathbb{C}\times\mathbb{C}$-module. This is not free (it is not zero, and it is not free of rank $\geq 1$). Thus, $M$ is locally free but not free.

## 3.4 Localisation as a quotient

Let $U\subseteq R$ be a subset, $S\subseteq R$ be its multiplicative closure. Define

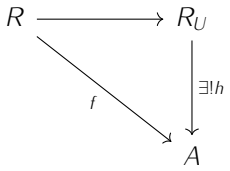
$$R_U=\frac{R[\{T_u : u\in U\}]}{\langle uT_u \mid u\in U\rangle}$$

Denote the ideal $I_U=\langle uT_u \mid u\in U\rangle$. Let $\overline{u},\overline{T_u}$ denote the images of $u,T_u$ respectively.

**Claim 3.4.1.** $R_U$ is isomorphic to $S^{-1}R$ as rings, and also as $R$-algebras. The isomorphism is given by

$$\begin{aligned}R_U &\leftrightarrow S^{-1}R\\ \overline{T_u} &\mapsto \frac{1}{u}\\ \overline{r}\,\overline{T_{u_1}}\cdots\overline{T_{u_n}} &\leftarrow\!\shortmid \frac{r}{u_1\cdots u_n}\end{aligned}$$

*Proof.* We will show that $R_U$ satisfies the universal property of localisation. Let $A$ be any ring, $f:R\to A$ any ring homomorphism, sending $U$ to units.




Since $A$ is an $R$-algebra via $f$, the diagram commutes if and only if $h$ is an $R$-algebra as well. But we have the bijection

$$\mathrm{Hom}_{R-\mathrm{alg}}(R_U,A)\leftrightarrow\{\varphi:U\to A \mid f(u)\varphi(u)=1\}$$

But the set on the right hand side has one elmeent. □

**Example 3.4.2**

For $x\in R$, we can invert $x$, and we have that

$$R_x\cong\frac{R[t]}{\langle tx-1\rangle}$$

Lecture 13

# 4 Nakayama's lemma

**Proposition 4.0.1** (Cayley-Hamilton). Let $M$ be a finitely generated $R$-module, $f : M \to M$ an $R$-linear map, $mf a \trianglelefteq R$ an ideal, with $f(M) \subseteq \mathfrak{a}M$. Then

$$f^n + a_1 f^{n-1} + a_n \,\mathrm{id} = 0$$

where $a_i \in \mathfrak{a}$.

*Proof.* Say $M = \mathrm{span}_R\{m_1, \ldots, m_n\}$, then $\mathfrak{a}M = \mathrm{span}_{\mathfrak{a}}\{m_1, \ldots, m_n\}$. Therefore,

$$\begin{pmatrix} f(m_1) \\ \vdots \\ f(m_n) \end{pmatrix} = P \begin{pmatrix} m_1 \\ \vdots \\ m_n \end{pmatrix}$$

where $P \in \mathrm{Mat}_n(\mathfrak{a})$. Take $\rho : R \to \mathrm{End}(M)$ to be the structure ring homomorphism of $M$ as an $R$-module, then we can define

$$\begin{aligned} R[t] &\to \mathrm{End}_R(M) \\ t &\mapsto f \end{aligned}$$

which makes $M$ into an $R[t]$-module. Using this,

$$t \cdot \begin{pmatrix} m_1 \\ \vdots \\ m_n \end{pmatrix} = P \begin{pmatrix} m_1 \\ \vdots \\ m_n \end{pmatrix}$$

and so

$$Q \begin{pmatrix} m_1 \\ \vdots \\ m_n = 0 \end{pmatrix}$$

where $Q = t \cdot I_n - P = 0$. Multiplying by $\mathrm{adj}(Q)$, we get that

$$\det(Q) \begin{pmatrix} m_1 \\ \vdots \\ m_n \end{pmatrix} = 0$$

Hence $\det(Q)m = 0$ for all $m \in M$, and so $m \mapsto \det(Q)m$ is the zero map. But then $\det(Q)$ gives the polynomial as required. □

**Corollary 4.0.2.** Let $M$ be a finitely generated $R$-module. $\mathfrak{a} \trianglelefteq R$ an ideal, if $\mathfrak{a}M = M$, then there exists $a \in \mathfrak{a}$ such that $am = m$ for every $m \in M$.

*Proof.* Apply Cayley-Hamilton with $f = \mathrm{id}_M$, we get that

$$(1 + a_1 + \cdots + a_n)\,\mathrm{id}_M = 0$$

and so we can take $a = -(a_1 + \cdots + a_n)$. □

**Definition 4.0.3** (Jacobson radical)
The *Jacobson radical* of a ring $R$ is

$$J(R) = \bigcap_{\mathfrak{m} \trianglelefteq R \text{ maximal}} \mathfrak{m}$$

Example 4.0.4

If $(R, \mathfrak{m})$ is a local ring, then $J(R) = \mathfrak{m}$. On the other hand, $J(\mathbb{Z}) = 0$.

**Proposition 4.0.5.** For $x \in R$, $x \in J(R)$ if and only if $1 - xy$ is a unit in $R$ for every $y \in R$.

*Proof.* Suppose that $x \in J(R)$, and suppose for contradiction that $1 - xy$ is not a unit, for some $y \in R$. With this, $1 - xy$ is contained in a maximal ideal $\mathfrak{m}$. Since $x \in J(R)$, $x \in \mathfrak{m}$. Thus,

$$1 = (1 - xy) + xy \in \mathfrak{m}$$

Contradiction. On the other hand, if $x \notin J(R)$, then there exists a maximal ideal $\mathfrak{m}$ such that $x \notin \mathfrak{m}$. Then $\mathfrak{m} + \langle x \rangle = R$. In particular, there exists $t \in \mathfrak{m}, y \in R$ such that $t + xy = 1$. In this case, $1 - xy = t \in \mathfrak{m}$, and so it is not a unit. □

**Proposition 4.0.6** (Nakayama's lemma)**.** Let $M$ be a finitely generated $R$-module, $\mathfrak{a} \leq J(R)$ is an ideal of $R$, with $\mathfrak{a}M = M$. Then $M = 0$.

*Proof.* By corollary 4.0.2, there exists $a \in \mathfrak{a}$ such that $am = m$ for all $m \in M$. By proposition 4.0.5, $1 = a$ is a unit, and so we can multiply by $(1 - a)^{-1}$, to get that

$$m = (1 - a)^{-1}(1 - a)m = (1 - a)^{-1} \cdot 0 = 0$$

□

**Corollary 4.0.7.** Let $M$ be a finitely generated $R$-module, $N \leq M$ an $R$-submodule, $\mathfrak{a} \leq J(R)$ an ideal, such that

$$N + \mathfrak{a}M = M$$

then $N = M$.

*Proof.*

$$\mathfrak{a} \cdot \left( \frac{M}{N} \right) = \frac{\mathfrak{a}M + N}{N} = \frac{M}{N}$$

Therefore, by Nakayama, $M/N = 0$, and so $N = M$. □

# 5 Integral and finite extensions

Definition 5.0.1 (integral)

Let $A$ be an $R$-algebra, $x \in A$ is *integral over* $R$ if there exists $f \in R[t]$ monic, such that $f(x) = 0$.

Example 5.0.2

If $K$ is a field, $A$ is a $K$-algebra, $x \in A$, then $x$ is integral over $K$ if and only if it is algebraic over $K$.

Example 5.0.3

We will see later

1. the elements of $\mathbb{Q}$ which are integral over $\mathbb{Z}$ is just $\mathbb{Z}$,
2. the $\mathbb{Z}$ integral elements of $\mathbb{Q}(\sqrt{2})$ is $\mathbb{Z}[\sqrt{2}]$,

3. the $\mathbb{Z}$ integral elements of $\mathbb{Q}(\sqrt{5})$ is $\mathbb{Z}\left[\frac{1+\sqrt{5}}{2}\right]$

To see this, we can also recall Part II Number Fields and the ring of integers of a number field.

**Definition 5.0.4** (faithful)

An $R$-module $M$ is *faithful* if the structure ring homomorphism $R \to \mathrm{End}_R(M)$ is injective.
That is, for every non-zero $r \in R$, there exists $m \in M$ such that $rm \neq 0$.

**Example 5.0.5**

Let $R \subseteq A$ be rings, and so $A$ is an $R$-module in a natural way. It must be faithful, since we have $r1 = r$.

**Proposition 5.0.6.** Let $R \subseteq A$ be rings, $x \in A$. Then $R[x] \subseteq A$ is a subring, which makes $A$ into an $R[x]$-algebra (and thus an $R[x]$-module). Then $x$ is $R[x]$-integral if and only if there exists $M \subseteq A$ such that

1. $M$ is a faithful $R[x]$-module, that is, $M$ is an $R$-submodule of $A$, $xM \subseteq M$, and $R[x] \to \mathrm{End}_{R[x]}(M)$ is injective,
2. $M$ is finitely generated as an $R$-module.

*Proof.* Suppose such an $M$ exists. With this, we have an $R$-linear map $f : M \to M$,

$$f(m) = xm$$

Since $M$ is a finitely generated $R$-module, we can apply Cayley-Hamilton (proposition 4.0.1), to get

$$f^n + r_1 f^{n-1} + \cdots + r_n = 0$$

where $r_i \in R$. Evaluating at $m \in M$, we get that

$$(x^n + r_1 x^{n-1} + \cdots + r_n)(m) = 0$$

Since $M$ is a faithful $R[x]$-module, $x^n + f_1 x^{n-1} + \cdots + r_n = 0$ That is, $x$ is integral over $R$. Now suppose $x$ is integral over $R$. Then

$$x^n + r_1 x^{n-1} + \cdots + r_n = 0$$

for some $r_i \in R$. Take

$$M = \mathrm{span}_R\{1, x, \cdots, x^{n-1}\}$$

satisfies $xM = M$, and as $1 \in M$, it is faithful. The fact that it is finitely generated is clear by definition. □

Lecture 14

**Definition 5.0.7** (integral)

Let $A$ be an $R$-algebra. Then $A$ is *integral over* $R$ if every $x \in A$ is integral over $R$.

**Definition 5.0.8** (finite over)

Let $A$ be an $R$-algebra, then $A$ is *finite over* $A$ if it is finitely generated as an $R$-module.

**Proposition 5.0.9.** Let $A$ be an $R$-algebra. Then the following are equivalent:

(i) $A$ is a finitely generated integral $R$-algebra,

(ii) $A$ is generated as an $R$-algebra by a finite set of integral elements,

(iii) $A$ is finite over $R$,

*Proof.* (i) $\implies$ (ii) is trivial. Suppose (ii) holds. Then $A$ is generated by $\alpha_1, \dots, \alpha_m$ as an $R$-algebra. But $\alpha_i$ being integral implies that

$$\alpha_i^{n_i} + r_{i,1}\alpha_i^{n_i-1} + \cdots + r_{i,n_i} = 0$$

That is,

$$\alpha_i^{n_i} \in \operatorname{span}_R\{1, \alpha_i, \dots, \alpha_i^{n_i-1}\}$$

But this means that for all $e_1, \dots e_n \geq 0$,

$$\alpha_1^{e_1} \cdots \alpha_m^{e_m} \in \operatorname{span}_R\{\alpha_1^{f_1} \cdots \alpha_m^{f_m} \mid 0 \leq f_i \leq n_i - 1\}$$

Hence $A$ is a finitely generated $R$-module.

Finally, suppose (iii) holds. If $A$ is finitely generated as an $R$-module, then it is necessarily finitely generated as an $R$-algebra. Choose $\alpha \in A$, we would like to show that $\alpha$ is integral over $R$. Let $\rho; R \to A$ be the structure ring homomorphism of $A$ as an $R$-algebra. Then $\rho(R)$ is a subring of $A$. With this, it then makes sense to consider $\rho(R)[\alpha]$ as a subring of $A$.

Next, $A$ is a $\rho(R)[\alpha]$-module, and it must be faithful as $1 \in A$. Using this, and the fact that $A$ is a finitely generated $\rho(R)[\alpha]$-module, so by proposition 5.0.6, $\alpha$ is integral over $\rho(R)$. Equivalently, $\alpha$ is integral over $R$. □

**Proposition 5.0.10.** If $A$ is an $R$-algebra, $\mathcal{O}$ is the integral elements of $A$, then $\mathcal{O}$ is an $R$-subalgebra of $A$.

*Proof.* Take $x, y \in \mathcal{O}$. Then this is a finite set of $R$-integral elements, and so must generate an integral $R$-subalgebra of $A$. But this contains $x \pm y, xy$, which must then be integral. Hence $\mathcal{O}$ is a ring. The fact that it is an $R$-subalgebra is clear. □

**Proposition 5.0.11.** If $A \subseteq B \subseteq C$ are rings,

(i) if $C$ is finite over $B$, and $B$ is finite over $A$, then $C$ is finite over $A$.

(ii) if $C$ is integral over $B$, $B$ is integral over $A$, then $C$ is integral over $A$.

*Proof.* For (i), if $C = \operatorname{span}_B\{\gamma_1, \dots, \gamma_n\}$, $B = \operatorname{span}_A\{\beta_1, \dots, \beta_\ell\}$, then $C = \operatorname{span}_A\{\beta_i\gamma_j\}$.

For (ii), let $c \in C$. We would like to show that $c$ is $A$-integral. We know that $c$ is $B$-integral, and so $f(c) = 0$ for some

$$f(T) = T^n + b_1T^{n-1} + \cdots + b_n \in B[T]$$

Hence $f \in A[b_1, \dots, b_n][T]$. Set $A' = A[b_1, \dots, b_n]$. Then we have inclusions

$$A \subseteq A' \subseteq A'[c]$$

Both inclusions are integral, as they are generated by finitely many integral elements. But this tells us that both extensions are finite by proposition 5.0.9. By (i), $A \subseteq A'[c]$ is finite, and so $A \subseteq A'[c]$ is integral, and so $c$ is integral over $A$. □

**Definition 5.0.12**

Let $A \subseteq B$ be rings. The *integral closure* of $A$ in $B$ is

$$\overline{A} = \{b \in B \mid b \text{ integral over } A\}$$

We say that $A$ is *integrally closed* if $A = \overline{A}$.

If $A$ is an integral domain, then its *integral closure* is its integral closure in $\mathrm{Frac}(A)$, and it is *integrally closed* if it is integrally closed in $\mathrm{Frac}(A)$.

**Example 5.0.13**

Consider $A = \mathbb{Z}[\sqrt{5}]$. This is not integrally closed, since $\mathrm{Frac}(A) = \mathbb{Q}(\sqrt{5})$. In this case,

$$\alpha = \frac{1+\sqrt{5}}{2} \in \mathrm{Frac}(A) \setminus A$$

But $\alpha$ is integral over $A$, since $\alpha^2 - \alpha - 1 = 0$.

**Example 5.0.14**

$\mathbb{Z}$ and $k[t_1, \cdots, t_n]$ are integrally closed.

**Proposition 5.0.15.** If $A$ is a UFD, then $A$ is integrally closed.

*Proof.* Take $x \in \mathrm{Frac}(A) \setminus A$, say $x = a/b$, $a, b \in A$, with some $p \in A$ prime, $p \mid b$ but $p \nmid a$. If $x$ is $A$-integral, then

$$\left(\frac{a}{b}\right)^n + a_1 \left(\frac{a}{b}\right)^{n-1} + \cdots + a_0 = 0$$

Multiply through by $b^n$, we get

$$a^n = -b(a_1 + a_2 b + \cdots + a_n b^{n-1})$$

Since $p \mid b$, $p$ divides the right hand side, and so $p \in a^n$. Thus, $p \mid a$. □

**Lemma 5.0.16.** If $A \subseteq B$ are rings, $\overline{A}$ the integral closure of $A$ in $B$, then $\overline{A}$ is integrally closed over $A$.

*Proof.* If $x \in B$ is integral over $\overline{A}$, then we have integral extensions

$$A \subseteq \overline{A} \subseteq \overline{A}[x]$$

By transitivity, $A \subseteq \overline{A}[x]$ is integral, and so $x$ is integral over $A$, that is, $x \in \overline{A}$. □

**Proposition 5.0.17.** Let $A \subseteq B$ be rings,

(i) If $B$ is integral over $A$,

(a) for every ideal $\mathfrak{b}$ of $B$,

$$\frac{B}{\mathfrak{b}} \text{ is integral over } \frac{A}{\mathfrak{b} \cap A}$$

(b) if $S \subseteq A$ is a multiplicative set, then $S^{-1}B$ is integral over $S^{-1}A$,

(ii) If $\overline{A}$ is the integral closure of $A$ in $B$, then then $S^{-1}\overline{A}$ is the integral closure of $S^{-1}A$ in $S^{-1}B$. That is, $\overline{S^{-1}A} = S^{-1}\overline{A}$

*Proof.* See notes. □

**Lemma 5.0.18.** Suppose $A \subseteq B$ is an integral extension of rings,

(i) $A \cap B^\times = A^\times$,

(ii) if $A, B$ are domains, then $A$ is a field if and only if $B$ is a field.

*Proof.* For (i), $\supseteq$ is clear. Conversely, take $a \in A \cap B^\times$. Then there exists $b \in B$ such that $ab = 1$. We need to show that $b \in A$. We know that $b$ is integral over $A$, that is,

$$b^n + a_1 b^{n-1} + \cdots + a_n = 0$$

Multiply this by $a^{n-1}$, we get

$$b + a_1 + a_2 a + \cdots + a_n a^{n-1} = 0$$

But $a_1 + a_2 a + \cdots + a_n a^{n-1} \in A$, and so $b \in A$.

For (ii), suppose that $B$ is a field. Then

$$A^\times = A \cap B^\times = A \cap (B \setminus \{0\}) = A \setminus \{0\}$$

and so $A$ is a field. Now suppose $A$ is a field. Let $b \in B$ be non-zero. Since $b$ is integral over $A$,

$$b^n + a_1 b^{n-1} + \cdots + a_n = 0$$

where $n$ is *minimal*. With this,

$$b(\underbrace{b^{n-1} + a_1 b^{n-2} + \cdots + a_{n-1}}_{=\delta}) = -a_n$$

By minimality, $\delta \neq 0$. Therefore, $a_n \neq 0$ as it is a domain. But $a_n \in A$ is a unit, so

Lecture 15

$$b(a_n^{-1}\delta) = 1$$

and so $b$ is a unit. □

**Corollary 5.0.19.** Let $A \subseteq B$ be an integral extension of rings, $\mathfrak{q}$ a prime ideal of $B$. Then $\mathfrak{q}$ is a maximal ideal of $B$ if and only if $\mathfrak{q} \cap A$ is a maximal ideal of $A$.

*Proof.* We have a ring embedding

$$\frac{A}{\mathfrak{q} \cap A} \hookrightarrow \frac{B}{\mathfrak{q}}$$

and these are integral domains as $\mathfrak{q}$ is prime. Moreover, this is an integral extension, and so we are done. □

# 6 Noether normalisation and Hilbert's Nullstellensatz

## 6.1 Noether normalisation

Throughout, let $k$ be a field.

**Definition 6.1.1** (algebraically independent)

If $A$ is a $k$-algebra, and $x_1, \ldots, x_n \in A$, then $x_1, \ldots, x_n$ are *$k$-algebraically independent* if for every $p \in k[T_1, \ldots, T_n]$ non-zero, $p(x_1, \ldots, x_n) \neq 0$. That is, the $k$-algebra homomorphism $k[T_1, \ldots, T_n] \to A$ given by sending $T_i$ to $x_i$ is injective.

**Theorem 6.1.2** (Noether normalisation). If $A \neq 0$ is a finitely generated $k$-algebra, then there exists $x_1, \ldots, x_n \in A$, which are $k$-algebraically independent, such that $A$ is finite over

$$A' = k[x_1, \ldots, x_n]$$

**Example 6.1.3** (of the method of proof)

Let $A = k[t, t^{-1}]$. First of all, note that $k[t] \subseteq k[t, t^{-1}]$ is not a finite extension. To see this, suppose it was, then $t^{-1}$ is integral over $k[t]$. That is,

$$t^{-n} \in \text{span}_{k[t]}\{1, t^{-1}, \dots, t^{-(n-1)}\}$$

Multiply through by $t^n$, we get

$$1 \in \text{span}_{k[t]}\{t^n, t^{n-1}, \dots, t\}$$

which is a contradiction. However, let $c \in k$ (which we will choose later). Then

$$A = k[t, t^{-1}] = k[t, t^{-1} - ct]$$

**Claim 6.1.4.** $k[T^{-1} - cT] \subseteq A$ is a finite extension for "most" c.

*Proof.* Since $tt^{-1} - 1 = 0$, we have that

$$((t^{-1} - ct) + ct)t - 1 = 0$$

Expanding,

$$ct^2 + (t^{-1} - ct)t - 1 = 0$$

Thus, if $c \neq 0$, then we can divide by $c$ to show that $t$ is integral over $k[t - ct^{-1}]$. □

*Proof of theorem 6.1.2 assuming k is infinite.* We will induct on the minimal number $m$ of generators of $A$ as an $k$-algebra.

**Base case:** $m = 0$ is trivial since $A = k$. We can take $A' = A$.

**Inducive step:** Suppose $A$ is generated by $x_1, \dots, x_m \in A$ as an $k$-algebra. If $x_1, \dots, x_m$ are algebraically independent, then we can take $A = A'$. Otherwise,

**Claim 6.1.5.** There exists $c_1, \dots, c_{m-1} \in k$ such that $x_m$ is integral over

$$B = k[x_1 - c_1 x_m, \dots, x_{m-1} - c_{m-1} x_m]$$

Assuming the claim, then $A = B[x_n]$, and so $A$ is finite over $B$. But $B$ is generated by $m - 1$ elements, and so by induction, $B$ contains $z_1, \dots, z_n \in B$, with $B$ finite over $A' = k[z_1, \dots, z_n]$. Then $A$ is finite over $A'$ by transitivity.

*Proof of claim 6.1.5.* Since $x_1, \dots, x_n$ are not algebraically independent over $k$, there exists a non-zero $f \in k[t_1, \dots, t_m]$, with

$$f(x_1, \dots, x_m) = 0$$

We would like to prove that $x_m$ is integral over $B$, where $c_i \in k$ we will choose later. Write

$$f = \sum_{i=0}^{r} f_{[i]}$$

as a sum of homogeneous parts. Set $F = f_{[r]}$ for the highest order part. For $c_1, \dots, c_{m-1} \in k$, set

$$g(t_1, \dots, t_m) = f(t_1 + c_1 t_m, \dots, t_{m-1} + c_{m-1} t_m, t_m) = F(c_1, \dots, c_{m-1}, 1)t_m^r + h(t_1, \dots, t_m)$$

where each term in $h$ has degree of $t_m$ less than $r$. Note

$$g(x_1 - c_1 x_m, \cdots, x_{m-1} - c_{m-1} x_m, x_m) = f(x_1, \dots, x_m) = 0$$

and that $g$ as a polynomial in $t_m$ over $k[t_1, \dots, t_{m-1}]$ has degree at most $r$, and the coefficient of $t_m^r$ is $F(c_1, \dots, c_{m-1}, 1)$, Since $F(t_1, \dots, t_m)$ is a non-zero homogeneous polynomial, and so $F(t_1, \dots, t_{m-1}, 1)$ is not zero. Therefore, there are $c_1, \dots, c_{m-1}$, with

$$F(c_1, \dots, c_{m-1}) \neq 0$$

since we are working over an infinite field (Schwartz-Zippel). □

□

**Remark 6.1.6.** Noether normalisation is true for any field.

From the example

$$k[t, t^{-1}] \cong \frac{k[x, y]}{\langle xy - 1 \rangle}$$

Geometrically, $xy - 1$ is a hyperbola. The projection onto the $x$-axis is not surjective, but the projection onto $y = cx$ is surjective for $c \neq 0$.

## 6.2 Hilbert Nullstellensatz

**Proposition 6.2.1** (Zariski's lemma). Let $k \subseteq L$ be fields, with $L$ finitely generated as a $k$-algebra. Then $\dim_k(L) < \infty$.

*Proof.* By Noether normalisation, we have a finite extension $k[x_1, \dots, x_\ell] \leq L$ where the $x_i$ are algebraically independent. Moreover, this is an integral extension, and so $k[x_1, \dots, x_\ell]$ is a field. So $\ell = 0$. Hence $k \leq L$ is a finite extension. □

Lecture 16

From now on, fix a field extension $\Omega/k$, where $\Omega$ is algebraically closed.

**Definition 6.2.2** (vanishing locus, algebraic set)
For $S \subseteq k[T_1, \dots, T_n]$, define

$$\mathbb{V}(S) = \{x \in \Omega^n \mid f(x) = 0 \text{ for all } f \in S\}$$

we call such sets *$k$-algebraic sets*

**Definition 6.2.3** (ideal of a subset)
For $X \subseteq \Omega^n$, define

$$I(X) = \{f \in k[T_1, \dots, T_n] \mid f(x) = 0 \text{ for all } x \in X\} \trianglelefteq k[T_1, \dots, T_n]$$

**Remark 6.2.4.** Note $\mathbb{V}(S) = \mathbb{V}(\langle S \rangle)$.

Recall from field theory that if $L/k$ is a finite field extension, then there exists a $k$-homomorphism $L \to \Omega$.

**Theorem 6.2.5.** Let $\mathfrak{a} \trianglelefteq k[T_1, \dots, T_n]$ be an ideal. Then

(i) (Weak Nullstellensatz) $\mathbb{V}(\mathfrak{a}) = \varnothing$ if and only if $1 \in \mathfrak{a}$,

(ii) (Strong Nullstellensatz) $I(\mathbb{V}(\mathfrak{a})) = \sqrt{\mathfrak{a}}$.

*Proof.* For (i), $\Longleftarrow$ is clear. Now suppose $1 \notin \mathfrak{a}$. Hence there exists a maximal ideal $\mathfrak{m}$ of $k[T_1, \ldots, T_n]$ containing $\mathfrak{a}$, and so $L = k[T_1, \ldots, T_n]/\mathfrak{m}$ is a field, and it is also finitely generated as a $k$-algebra. By Zariski's lemma, $\dim_k(L) < \infty$. Hence there exists a $k$-homomorphism $L \to \Omega$.

Consider the composition $\varphi : k[T_1, \ldots, T_n] \to L \to \Omega$. In this case, $\ker(\varphi) = \mathfrak{m}$. Define

$$x = (\varphi(T_1), \ldots, \varphi(T_n)) \in \Omega^n$$

Then for $f \in k[T_1, \ldots, T_n]$,

$$\varphi(f) = f(\overline{x})$$

Hence for all $f \in \mathfrak{a} \subseteq \mathfrak{m}$,

$$f(\overline{x}) = \varphi(f) = 0$$

For (ii), let $f \in \sqrt{\mathfrak{a}}$. Then then $f^\ell \in \mathfrak{a}$ for some $\ell$, and thus $f^\ell(x) = 0$ for all $x \in \mathbb{V}(\mathfrak{a})$. But we are working in a field, and so $f(x) = 0$ for all $x \in \mathbb{V}(\mathfrak{a})$, i.e. $f \in I(\mathbb{V}(\mathfrak{a}))$.

Conversely, take $f \in I(\mathbb{V}(\mathfrak{a}))$. We want to show that $f \in \sqrt{\mathfrak{a}}$. Equivalently, $\overline{f}$ is nilpotent in $R = k[T_1, \ldots, T_n]/\mathfrak{a}$. In turn, this is equivalent to

$$R_{\overline{f}} = 0$$

But recall that

$$R_{\overline{f}} = \frac{R[T_1, \ldots, T_n, U]}{\mathfrak{a}^e + \langle Uf - 1 \rangle}$$

Let $\mathfrak{b} = \mathfrak{a}^e + \langle UF - 1 \rangle$. Hence we need to show that $1 \in \mathfrak{b}$. By the Weak Nullstellensatz, it suffices to show $\mathbb{V}\mathfrak{b} = \varnothing$.

Take $x = (x_1, \ldots, x_n, u) \in \mathbb{V}(\mathfrak{b}) \subseteq \Omega^{n+1}$. Let $x' = (x_1, \ldots, x_n)$, then

$$x' \in \mathbb{V}(\mathfrak{a})$$

Hence $f(x')$, since $f \in I(\mathbb{V}(\mathfrak{a}))$. Considering the canonical embedding $k[T_1, \ldots, T_n] \hookrightarrow k[T_1, \ldots, T_n, U]$, $f(x') = 0$. Now $(Uf - 1)(x) = -1 \neq 0$, contradiction, as $Uf - 1 \in \mathfrak{b}$. □

Recall $\sqrt{\sqrt{I}} = \sqrt{I}$, and we have that

1. if $X \subseteq Y \subseteq \Omega^n$, then $I(Y) \subseteq I(X)$,
2. if $S \subseteq T \subseteq k[T_1, \ldots, T_n]$, then $\mathbb{V}(T) \subseteq \mathbb{V}(S)$,
3. if $S \subseteq k[T_1, \ldots, T_n]$, then $S = I(\mathbb{V}(S))$,
4. if $X \subseteq \Omega^n$, then $X \subseteq \mathbb{V}(I(X))$.
5. if $X \subseteq \Omega^n$ is an algebraic set, then $X = \mathbb{V}(I(X))$. This follows from writing $X = \mathbb{V}(\mathfrak{a})$.
6. if $X \subseteq \Omega^n$, then $I(X)$ is a radical ideal.

**Proposition 6.2.6.** We have a bijection

$$\begin{aligned} \{k\text{-alg. subsets of } \Omega^n\} &\leftrightarrow \{\text{radical ideals in } k[T_1, \ldots, T_n]\} \\ X &\mapsto I(X) \\ \mathbb{V}(\mathfrak{a}) &\hookleftarrow \mathfrak{a} \end{aligned}$$

*Proof.* We know $I(X)$ is radical, and $X = \mathbb{V}(I(X))$. Now take $\mathfrak{a} \in k[T_1, \ldots, T_n]$ a radical ideal, then by the strong Nullstellensatz

$$I(\mathbb{V}(\mathfrak{a})) = \sqrt{\mathfrak{a}} = \mathfrak{a}$$

□

**Remark 6.2.7.** Note that we defined algebraic subsets with respect to $k \subseteq \Omega$.

**Corollary 6.2.8.** Under the above correspondence, maximal ideals correspond to minimal non-empty algebraic sets. In particular, let $k = \Omega$ be an algebraically closed field. Then we have a bijection

$$\Omega^n \leftrightarrow \{\text{maximal ideals of } \Omega[T_1, \dots, T_n]\}$$
$$x = (x_1, \dots, x_n) \mapsto \mathfrak{m}_x = (T_1 - x_1, \dots, T_n - x_n)$$

*Proof.* The first part is just the fact that $\mathbb{V}$ and $I$ are order reversing.

Since $\Omega[T_1, \dots, T_n]/\mathfrak{m}_x = \Omega$, $\mathfrak{m}_x$ is a maximal ideal. Moreover, $\mathfrak{m}_x$ is the ideal of polynomials which vanish on $x$. To see this,

$$\mathfrak{m}_x \subseteq I(\{x\})$$

But $\mathfrak{m}_x$ is maximal, and $I(\{x\})$ is a proper ideal, and so equality holds. Moreover, $\mathbb{V}(\mathfrak{m}_x) = \{x\}$. The claim follows from the inclusion reversing bijection from before. □

Note that the requirement that $k = \Omega$ above is necessary. Consider the field extension $\mathbb{C}/\mathbb{R}$. In $\mathbb{R}[t]$, $\langle t^2 + 1 \rangle$ is a maximal ideal, but it corresponds to the points $\{i, -i\} \subseteq \mathbb{C}$. In general, for $\Omega/k$ as above, each point $x \in k^n$ is a minimal $k$-algebraic subsets of $\Omega^n$, but there can be more. If $\mathrm{char}(k) = 0$, then $x \in \Omega^n$ is $k$-algebraic if and only if the coordinates are in $k$. More generally, if $\Omega/k$ is separable.

On the other hand, if $k = \mathbb{F}_p(x)$ is the field of rational functions over $\mathbb{F}_p$, $\Omega = \overline{k}$, $n = 1$. Consider the polynomial

$$T^p - x \in k[T]$$

By Frobenius and that $k$ is algebraically closed, $T^p - x = (T - x^{1/p})^p$ over $\Omega$. Hence

$$\mathbb{V}(T^p - x) = \{x^{1/p}\}$$

Finally, note that every prime ideal is radical.

**Definition 6.2.9** (irreducible)
$X \subseteq \Omega^n$ is *irreducible* if $X$ is not the union $X = X_1 \cup X_2$, $X_1, X_2$ algebraic and $X \neq X_1, X_2$.

**Proposition 6.2.10.** Let $X \subseteq \Omega^n$ be an algebraic set. Then $X$ is irreducible if and only if $I(X)$ is prime.

*Proof.* See notes, or Part II Algebraic Geometry. □

# 7 Integral and finite extensions again

**Definition 7.0.1** (integral over an ideal)
If $A \subseteq B$, $\mathfrak{a} \trianglelefteq A$, $x \in B$ is *integral over* $\mathfrak{a}$ if

$$x^n + a_1 x^{n-1} + \cdots + a_n = 0$$

where $a_i \in \mathfrak{a}$.

Lecture 17

**Definition 7.0.2** (integral closure over an ideal)
If $A \subseteq B$ rings, $\mathfrak{a} \trianglelefteq A$, then the *integral closure of* $\mathfrak{a}$ *in* $B$ is

$$\{x \in B \mid x \text{ is } \mathfrak{a}\text{-integral}\}$$

**Proposition 7.0.3.** If $A \subseteq B$ are rings, $\overline{A}$ the integral closure of $A \subseteq B$, $\mathfrak{a} \subseteq A$ is an ideal. Then the integral closure of $\mathfrak{a}$ in $B$ is

$$\sqrt{\mathfrak{a}\overline{A}}$$

where we take the radical in $\overline{A}$.

*Proof.* Suppose $b \in B$ is $\mathfrak{a}$-integral, then

$$b^n + a_1 b^{n-1} + a_n = 0$$

with $a_i \in \mathfrak{a}$. In particular, $\mathfrak{b}$ is integral over $A$, and therefore, $b_0, \dots, b_{n-1} \in \overline{A}$. Using the above,

$$b^n \in \mathfrak{a}\overline{A}$$

and so $b \in \sqrt{\mathfrak{a}\overline{A}}$.

Now suppose $b \in \sqrt{\mathfrak{a}\overline{A}}$. Then $b^n \in \mathfrak{a}\overline{A}$ for some $n$, and so

$$b^n = \sum_{i=1}^{m} a_i x_i \tag{$*$}$$

where $a_i \in \mathfrak{a}$, $x_i \in \overline{A}$. Define the algebra

$$M := A[x_1, \dots, x_m]$$

Since each $x_i$ is integral over $A$, $M$ is a finite $A$-algebra. Moreover, from $(*)$, $b^n M \subseteq \mathfrak{a}M$. Now define $f : M \to M$,

$$f(m) = b^n m$$

This satisfies $f(M) \subseteq \mathfrak{a}M$, and $f$ is $A$-linear. Therefore, by Cayley-Hamilton,

$$f^\ell + \alpha_1 f^{\ell-1} + \cdots + \alpha_\ell = 0 \in \mathrm{End}_R(M)$$

where each $\alpha_i \in \mathfrak{a}$. Evaluating this at $1 \in A$, we get that

$$b^{n\ell} + \alpha_1 b^{n(\ell-1)} + \cdots + \alpha_\ell = 0 \in B$$

and so $b$ is $\mathfrak{a}$-integral. □

**Corollary 7.0.4.** Suppose $A \subseteq B$ are rings, $\mathfrak{a} \trianglelefteq A$, $b \in B$, then $b$ is $\mathfrak{a}$-integral if and only if $b$ is $\sqrt{\mathfrak{a}}$-integral.

*Proof.* By the proposition, it suffices to show

$$\sqrt{\mathfrak{a}\overline{A}} = \sqrt{\sqrt{\mathfrak{a}}\,\overline{A}}$$

$\subseteq$ is clear. For $\supseteq$, note that in general, $\sqrt{I}^e \subseteq \sqrt{I^e}$. Applying this to the above, we have that

$$\sqrt{\mathfrak{a}}\,\overline{A} \subseteq \sqrt{\mathfrak{a}\overline{A}}$$

and so

$$\sqrt{\sqrt{\mathfrak{a}}\,\overline{A}} \subseteq \sqrt{\mathfrak{a}\overline{A}}$$

□

**Proposition 7.0.5.** Let $A$ be an integrally closed[a] integral domain, and $A \subseteq B$ rings, $B$ is an integral domain, and an ideal $\mathfrak{a} \trianglelefteq A$. Let $b \in B$, We have a field extension $\mathrm{Frac}(B)/\mathrm{Frac}(A)$, and the following are equivalent:

(i) $b$ is integral over $\mathfrak{a}$

(ii) $b$ is algebraic over $\mathrm{Frac}(A)$, with minimal polynomial over $\mathrm{Frac}(A)$ of the form

$$T^n + a_1 T^{n-1} + \cdots + a_0$$

where $a_i \in \sqrt{\mathfrak{a}}$.

$a$ in $\mathrm{Frac}(A)$

*Proof.* Suppose (ii) holds, then $b$ is integral over $\sqrt{\mathfrak{a}}$ by definition. By the corollary, $b$ is integral over $\mathfrak{a}$.

Now suppose (i) holds. Let $F = \mathrm{Frac}(A)$. Then we have that

$$b^n + a_1 b^{n-1} + \cdots + a_n = 0$$

where $a_i \in \mathfrak{a}$. Set

$$h(T) = T^n + a_1 T^{n-1} + \cdots + a_n \in F[T]$$

Then $h(b) = 0$, and so $b$ is algebraic over $\mathrm{Frac}(A)$. Now let $f$ be the minimial polynomial of $b$ over $F$. Let $\Omega/F$ be an algebraically closed field. In this case,

$$f = \prod_{i=1}^{\ell} (T - \alpha_i) \tag{$*$}$$

where each $\alpha_i \in \Omega$. We would like to show that the coefficient of $f$ are in $\sqrt{\mathfrak{a}}$. Since $A$ is integrally closed, the integral closure of $\mathfrak{a}$ in $F$ is $\sqrt{\mathfrak{a}} \trianglelefteq A$. Thus, it suffices to show that the coefficients of $f$ are $\mathfrak{a}$-integral. Note that by definition, the coefficient of $f$ are in $F$.

Expanding $(*)$, we see the coefficients of $f$ are sums of products of the $\alpha_i$. By the proposition, the integral closure of $\mathfrak{a}$ in $\Omega$ is closed under sums and products (as it is an ideal). Therefore, we need to show that each $\alpha_i$ is integral over $A$.

In this case, $\alpha_i$ and $b$ have the same minimal polynomial over $\mathrm{Frac}(A)$, and therefore, there exists $\varphi_i : F(b) \to F(\alpha_i)$, which is a $F$-homomorphism, with $\varphi_i(b) = \alpha_i$. Since $h$ has coefficients in $F$,

$$h(\alpha_i) = h(\varphi_i(b)) = \varphi(h_i(b)) = 0$$

□

## 7.1 Cohen-Seidenberg theorems

Let $\iota : A \hookrightarrow B$ be the inclusion map. Then we have a pullback

$$\begin{aligned} i^* : \mathrm{Spec}(B) &\to \mathrm{Spec}(A) \\ \mathfrak{q}* &\mapsto \mathfrak{q} \cap A \end{aligned}$$

We are interested in studying $i^*$, in particular its fibres.

**Proposition 7.1.1** (incomparability). If $A \subseteq B$ is an integral extension, $\mathfrak{q}, \mathfrak{q}' \in \mathrm{Spec}(B)$, $\mathfrak{q} \subseteq \mathfrak{a}'$, and $\mathfrak{q} \cap A = \mathfrak{q}' \cap A$. Then $\mathfrak{q} = \mathfrak{q}'$.

That is, the elements of the fibres are pairwise incomparable.

*Proof.* Let $\mathfrak{p} = \mathfrak{q} \cap A = \mathfrak{q}' \cap A$, and $S = A \setminus \mathfrak{p}$. $\mathfrak{q}$ and $\mathfrak{q}'$ are prime ideals of $B$ not intersecting $S$, So

$$\mathfrak{q} = (S^{-1}\mathfrak{q})^c$$

where by $S^{-1}\mathfrak{q}$, we mean the extension of $\mathfrak{q}$ to $S^{-1}B$. Note this is not the localisation of $B$ at $\mathfrak{p}$, since $\mathfrak{p}$ need not be a prime in $B$. Similarly, $\mathfrak{q}' = (S^{-1}\mathfrak{q}')^c$. We would like to show that

$$S^{-1}\mathfrak{q} = S^{-1}\mathfrak{q}'$$

To see this,

$$S^{-1}\mathfrak{q} \cap A_{\mathfrak{p}} = S^{-1}\mathfrak{q} \cap S^{-1}A = S^{-1}(\mathfrak{q} \cap A) = S^{-1}\mathfrak{p} = \mathfrak{p}A_{\mathfrak{p}}$$

Similarly, $S^{-1}\mathfrak{q}' \cap A_{\mathfrak{p}} = \mathfrak{p}A_{\mathfrak{p}}$, which is the unique maximal ideal of $A_{\mathfrak{p}}$.

Since $A \subseteq B$ is an integral extension, so is $A_{\mathfrak{p}} \subseteq S^{-1}B$. Therefore, the contractions $S^{-1}\mathfrak{q}, S^{-1}\mathfrak{q}'$ are maximal ideals of $S^{-1}B$. But $\mathfrak{q} \subseteq \mathfrak{q}'$, and so they are equal. □

Lecture 18

**Proposition 7.1.2** (lying over). Let $A \subseteq B$ be an integral extension, $\mathfrak{p} \in \operatorname{Spec}(A)$. Then there exists $\mathfrak{q} \in \operatorname{Spec}(B)$ with $\mathfrak{q} \cap A = \mathfrak{p}$.

Equivalently, the natural map $\operatorname{Spec}(B) \to \operatorname{Spec}(A)$ is surjective.

*Proof.* Let $S = A \setminus \mathfrak{p}$, then we have the commutative diagram

$$\begin{array}{ccc} A & \longrightarrow & B \\ \downarrow & & \downarrow{\scriptstyle \beta} \\ A_{\mathfrak{p}} = S^{-1}A & \longrightarrow & S^{-1}B \end{array}$$

Take $\mathfrak{m} \in \operatorname{maxSpec}(S^{-1}B)$. Since $S^{-1}A \subseteq S^{-1}B$ is an integral extension, and so $\mathfrak{m} \cap S^{-1}A \in \operatorname{maxSpec}(S^{-1}A) = \{\mathfrak{p}A_{\mathfrak{p}}\}$. Hence $\mathfrak{m} \cap S^{-1}A = \mathfrak{p}A_{\mathfrak{p}}$. Under the localisation map, $\mathfrak{p}A_{\mathfrak{p}}$ contracts to $\mathfrak{p}$. Thus, $\mathfrak{m}$ contracts to $\mathfrak{p}$, and so $\mathfrak{q} = \beta^{-1}(\mathfrak{m})$ has $\mathfrak{q} \cap A = \mathfrak{p}$. □

**Proposition 7.1.3** (going up). Let $A \subseteq B$ be an integral extension of rings, let $\mathfrak{p}_1, \mathfrak{p}_2 \in \operatorname{Spec}(A)$, $\mathfrak{q}_1 \in \operatorname{Spec}(B)$, with $\mathfrak{p}_1 \subseteq \mathfrak{p}_2$, $\mathfrak{q}_1^c = \mathfrak{p}_1$. That is,

$$\begin{array}{ccc} \mathfrak{q}_1 & & \\ | & & \\ \mathfrak{p}_1 & \hookrightarrow & \mathfrak{p}_2 \end{array}$$

there exists $\mathfrak{q}_2 \in \operatorname{Spec}(B)$, with $\mathfrak{q}_1 \subseteq \mathfrak{q}_2$, and $\mathfrak{q}_2^c = \mathfrak{p}_2$. Note that in the diagram we use vertical line with no arrows to denote contraction.

*Proof.* $\mathfrak{p}_1 = \mathfrak{q}_1 \cap A$, and so we have an injective map $A/\mathfrak{p}_1 \to B/\mathfrak{q}_1$. This is an integral extension. From lying over, there exists a prime ideal $\mathfrak{q}_2/\mathfrak{q}_1 \in \operatorname{Spec}(B/\mathfrak{q}_1)$, with $\mathfrak{q}_2 \in \operatorname{Spec}(B)$, which contracts to $\mathfrak{p}_2/\mathfrak{p}_1 \in \operatorname{Spec}(A/\mathfrak{p}_1)$.

**Claim 7.1.4.** $\mathfrak{q}_2 \cap A = \mathfrak{p}_2$.

For this, consider the diagram

$$\begin{array}{ccc} A & \longrightarrow & B \\ \downarrow & & \downarrow \\ A/\mathfrak{p}_1 & \longrightarrow & B/\mathfrak{q}_1 \end{array}$$

Contracting along the bottom left we get $\mathfrak{p}_2$, and contracting along thr right gives $\mathfrak{q}_2$. □

**Proposition 7.1.5** (going down). Let $A \subseteq B$ be an integral extension of integral domains, and assume $A$ is integrally closed. Consider the diagram

$$\begin{array}{ccc} \mathfrak{q}_1 & & \\ | & & \\ \mathfrak{p}_1 & \hookleftarrow & \mathfrak{p}_2 \end{array}$$

Then there exists a prime $\mathfrak{q}_2 \in \operatorname{Spec}(B)$ with $\mathfrak{q}_2 \cap A = \mathfrak{p}_2$.

*Proof.* Consider the map

$$A \hookrightarrow B \hookrightarrow B_{\mathfrak{q}_1}$$

**Claim 7.1.6.** There exists $\mathfrak{n} \in \operatorname{Spec}(B_{\mathfrak{q}_1})$ such that $\mathfrak{n} \cap A = \mathfrak{p}_2$.

Assuming the claim, $(\mathfrak{n} \cap B) \cap A = \mathfrak{p}_2$, and $\mathfrak{n} \cap B$ is a prime ideal of $B$ contained in $\mathfrak{q}_1$.

To prove the claim, it suffices to show that

$$(\mathfrak{p}_2 B)B_{\mathfrak{q}_1} = \mathfrak{p}_2 B_{\mathfrak{q}_1} \cap A \subseteq \mathfrak{p}_2$$

Take $y/s \in (\mathfrak{p}_2 B)B_{\mathfrak{q}_1} \cap A$, with $y \in \mathfrak{p}_2 B$, $s \in B \setminus \mathfrak{q}_1$. Now $A \subseteq B$ is an integral extension, therefore the integral closure of $\mathfrak{p}_2$ in $B$ is $\sqrt{\mathfrak{p}_2 B}$. Thus, $y$ is integral over $\mathfrak{p}_2$. Since $A$ is integrally closed, by proposition 7.0.5, $y \in \operatorname{Frac}(A)$ is algebraic over $\operatorname{Frac}(A)$, and the minimal polynomial has the form

$$y^r + u_1 y^{r-1} + \cdots + u_r = 0$$

where $u_i \in \mathfrak{p}_2$ (note any prime ideal is radical). We can then write

$$y = \frac{y}{s} s$$

$y, s \in B \subseteq \operatorname{Frac}(B)$, $y/s \in A \subseteq \operatorname{Frac}(A)$, and so we have

$$\left(\frac{y}{s}s\right)^r + u_1 \left(\frac{y}{s}s\right)^{r-1} + \cdots + u_r = 0$$

Multiply through by $(s/y)^r$,

$$s^r + \frac{s}{y} u_1 s^{r-1} + \cdots + \left(\frac{s}{y}\right)^r u_r = 0 \tag{$*$}$$

This is the minimal polynomial of $s$ over $\operatorname{Frac}(A)$, since the process above is reversible. But $s \in B$, and so $s$ is integral over $A$. Therefore, the coefficients of $(*)$ must all be in $A$, again by proposition 7.0.5.

Suppose for contradiction $y/s \notin \mathfrak{p}_2$. Then

$$u_i = \left(\frac{y}{s}\right)^i \left(\frac{s}{y}\right)^i u_i$$

Then $(y/s)^i \in A \setminus \mathfrak{p}_2$, and we know that $(s/y)^i u_i \in A$. Since $u_i \in \mathfrak{p}_2$, we must have that $(s/y)^i u_i \in \mathfrak{p}_2$. With this, by $(*)$,

$$s^r \in \mathfrak{p}_2 B \subseteq \mathfrak{p}_1 B = (\mathfrak{q}_1 \cap A)B \subseteq \mathfrak{q}_1$$

Hence $s \in \mathfrak{q}_1$. Contradiction. □

# 8 Primary decomposition

**Definition 8.0.1** (primary ideal)

Let $I$ be an ideal of $R$, then $I$ is *primary* if $R/I$ is non-zero, and every zero divisor in $R/I$ is nilpotent.

**Remark 8.0.2.** Contrast this with $I$ being prime if $R/I$ is an integral domain, and $I$ is radical if $R/I$ has no non-zero nilpotent elements.

In particular, any prime ideal is radical and primary. Note $R$ is radical, but not prime nor primary.

**Example 8.0.3**

In $\mathbb{Z}$, $\langle 6 \rangle$ is radical, but not primary, since in $\mathbb{R}/6$, there are no non-zero nilpotent elements, but $2 \times 3 = 6$. But $\langle 9 \rangle$ is primary, but not radical.

# Index


algebra, 3
    finitely generated, 4
    homomorphism, 4
algebraically independent, 39
Artinian
    module, 2
    ring, 2

Cayley-Hamilton, 34
contraction of an ideal, 28

divisible group, 20

epimorphism, 24
exact sequence, 2
    short exact sequence, 2
extension of an ideal, 28
extension of scalars
    for morphisms, 15
    of algebras, 15
    of modules, 12

faithful, 36
finite
    extension of algebras, 36
flat module, 18

going down, 46
going up, 46

Hilbert basis theorem, 4
Hom functors, 16

ideal of a subset, 41
integral
    element, 35
    element over an ideal, 43
    extension, 36
integral closure, 37
    over an ideal, 43
integrally closed, 38
irreducible
    algebraic set, 43

Jacobson radical, 34

$k$-algebraic set, 41
Kronecker product, 9

left exact functor
    Hom, 17
local ring, 31
localisation
    of a module, 23
    of a ring, 23
locally free, 32
lying over, 46

module, 1
multiplicative closure, 22
multiplicative subset, 22

Nakayama's lemma, 35
Noether normalisation, 39
Noetherian
    algebra, 3
    module, 2
    ring, 2
Nullstellensatz, 41

primary ideal, 47

restriction of scalars, 12
right exact functor, 16
    $\mathrm{T}_M$, 17

spectrum, $\mathrm{Spec}(R)$, 25

tensor product
    of $R$-linear maps, 9
    of $R$-modules, 5
    of algebras, 10
torsion free, 19

universal property of
    localisation of a ring, 23
    tensor product, 5
    tensor product of algebras, 11